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Student Number



2025

Parramatta High School

TRIAL EXAMINATION

Physics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using a black/blue pen.
- Draw diagrams using a pencil.
- NESA-approved calculators may be used.
- A formula sheet, data sheet, and Periodic Table are provided at the back of this paper.
- For questions in Section II, **show all relevant working** in the questions involving calculations.

Total marks
100

Section I -20 marks (pages 2-9)

- Attempt Questions 1– 20
- Allow about 35 minutes for this part.

Section II -80 marks (pages 11 -28)

- Attempt Questions 21–32
- Allow about 2 hours and 25 minutes for this part

This paper **MUST NOT** be removed from the examination room.

Section I

20 marks

Attempt questions 1 – 20

Allow about 35 minutes for this part.

Use the multiple-choice answer sheet for Questions 1-20.

1. Which of the following is equivalent to the SI unit of magnetic field, tesla (T)?

(A) $\frac{\text{Ampere}}{\text{Newton}}$

(B) $\frac{\text{Newton}}{\text{Ampere}}$

(C) $\frac{\text{joules}}{\text{Ampere} \cdot \text{Newton}}$

(D) $\frac{\text{Newton}}{\text{Ampere} \cdot \text{meter}}$

2. Which of the following is one of Kepler's laws of planetary motion?

(A) The laws of physics are the same in all inertial frames of reference.

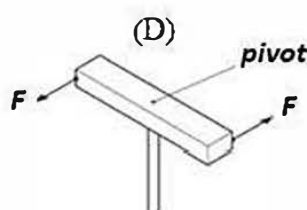
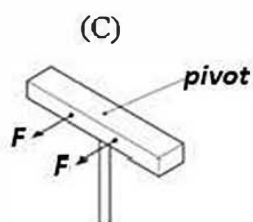
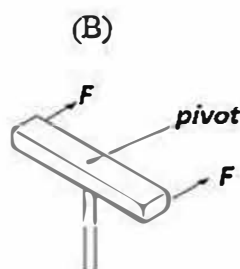
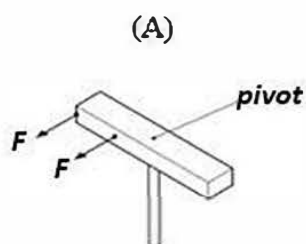
(B) All planets move about the Sun in elliptical orbits, having the Sun as one of the foci.

(C) The speed of light in a vacuum has the same value, c , in all inertial frames of reference.

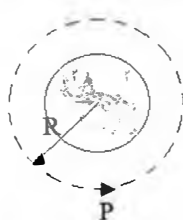
(D) The force of attraction between each pair of point particles is directly proportional to the product of their masses and inversely proportional to the square of the distance between them.

3. A wooden bar is pivoted at its centre so that it can rotate freely. Two equal forces F are applied to the bar.

In which diagram is the torque the greatest?



4. A spacecraft orbits Earth in a circular orbit of radius R , as shown below



When the spacecraft is at position P shown, a short burst of the ship's engines results in a small increase in its speed.

Which of the following diagrams shows the new orbit of the satellite (solid line)?

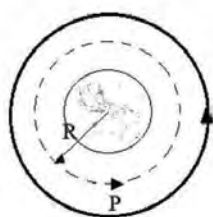
(A)



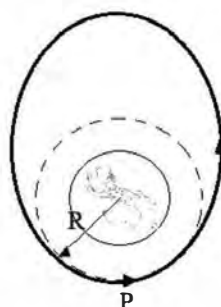
(B)



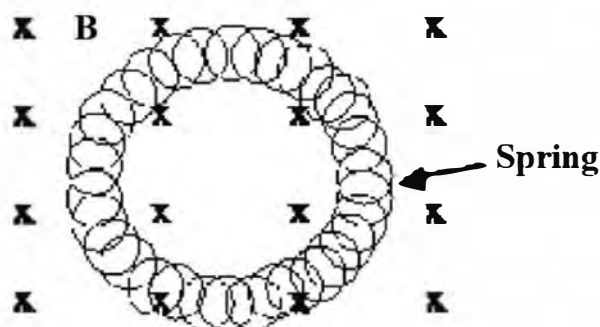
(C)



(D)



5. A metal spring has its ends attached so that it forms a circle. It is placed in a uniform magnetic field, as shown.

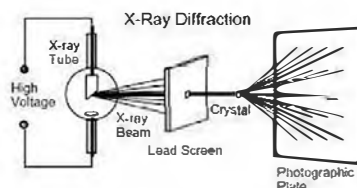


Which of the following will not cause a current to be induced in the spring?

- (A) Rotating the spring about a diameter
- (B) Changing the magnitude of the magnetic field
- (C) Moving the spring parallel to the magnetic field

(D) Moving the spring in and out of the magnetic field

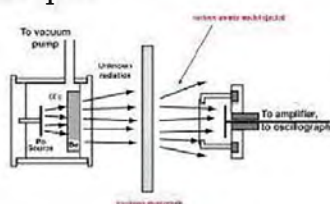
6. The following diagrams show X-ray diffraction of a crystalline solid. An interference pattern is incident on the photographic plate.



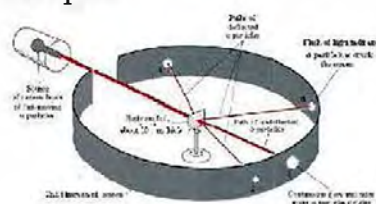
Which of the following causes the diffraction pattern?

- (A) The photographic plate that reacts to the X-ray beam.
 (B) The spacing between the atoms of the crystalline solid.
 (C) The way atoms are arranged in the crystalline structure.
 (D) The high-voltage X-ray beam issued from the X-ray tube.
7. The following shows two experimental setups.

Setup Y



Setup Z



Which of the following identifies the scientists who used each setup to develop their theory?

	Setup Y	Setup Z
(A)	Chadwick	Rutherford
(B)	Chadwick	Gieger-Marsden
(C)	Joliot-Curie	Gieger-Marsden
(D)	Joliot-Curie	Rutherford

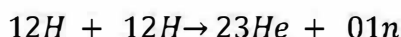
8. A diffraction grating has 10,000 slits ruled across 5 cm.

What is the wavelength of the light whose pair of third-order maxima subtend an angle of 90° ?

- (A) 1178 nm

- (B) 1666 nm
- (C) 4714 nm
- (D) 2654 nm

9. In our Sun, the following fusion process can occur.

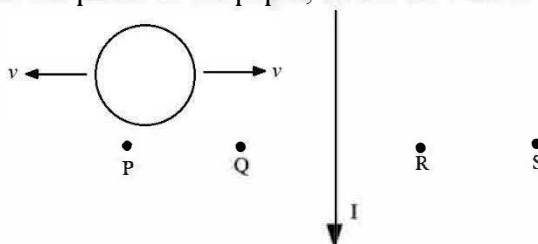


The table below shows the mass of the particles.

Particle	Mass (amu)
$01n$	1.008665
$12H$	2.014102
$23He$	3.016050

What is the energy in joules released during this fusion process?

- (A) 6.00×10^{-25}
 (B) 5.21×10^{-13}
 (C) 3.14×10^{14}
 (D) 1.89×10^{41}
10. A wire carries a constant current I down the page. A wire loop moves either to the right or left with a constant velocity v in the plane of the paper, as shown below.



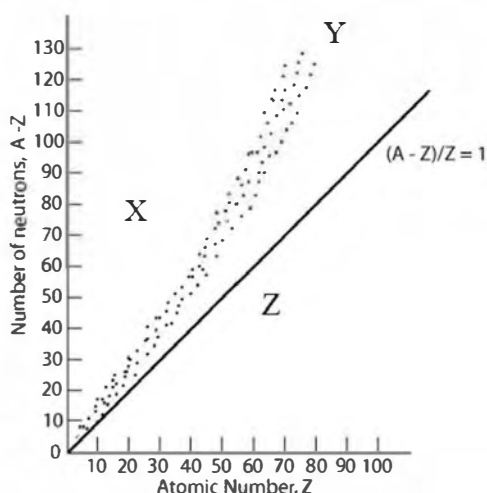
Which of the following statements is true?

- (A) Current flows clockwise as the loop moves from Q to P and clockwise as the loop moves from S to R.
 (B) Current flows counterclockwise as the loop moves from P to Q and clockwise as the loop moves from R to S.
 (C) Current flows clockwise as the loop moves from P to Q and counterclockwise as the loop moves from R to S.
 (D) Current flows counterclockwise as the loop moves from P to Q and counterclockwise as the loop moves from S to R
11. While conducting a photoelectric effect experiment, the stopping voltage of 1.25V was needed to reduce the current to zero.

What is the maximum velocity in m/s of the emitted photoelectrons?

- (A) 4.21×10^{-19}
 (B) 2.34×10^5
 (C) 6.63×10^5
 (D) 2.34×10^{30}

12. The graph shows the nuclear stability curve for stable isotopes and three regions labelled X, Y and Z.



Which row shows the most probable decay process for each region?

	<i>Region X</i>	<i>Region Y</i>	<i>Region Z</i>
(A)	$ZAX \rightarrow Z + 1AY + - 10$	$ZAX \rightarrow Z - 1AY ++ 10$	$ZAX \rightarrow Z - 2A - 4Y +$
(B)	$ZAX \rightarrow Z - 1AY ++ 10$	$ZAX \rightarrow Z - 2A - 4Y +$	$ZAX \rightarrow Z + 1AY + - 10$
(C)	$ZAX \rightarrow Z + 1AY + - 10$	$ZAX \rightarrow Z + 1AY + - 10$	$ZAX \rightarrow Z - 2A - 4Y +$
(D)	$ZAX \rightarrow Z + 1AY + - 10$	$ZAX \rightarrow Z - 2A - 4Y +$	$ZAX \rightarrow Z - 1AY ++ 10$

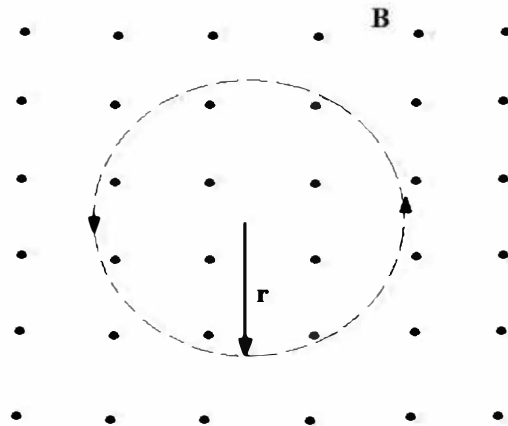
13. Hubble's constant is reported to be 82 km/s/Mpc, which corresponds to the age of the universe being 13.8 billion years old.

If a new measurement claims that the Hubble constant is about 164 km/s/Mpc and not 82 km/s/Mpc, then how old is the universe with this new Hubble constant?

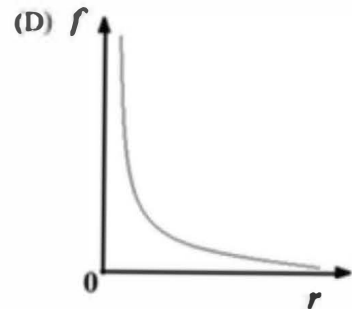
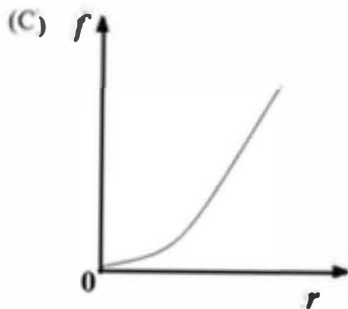
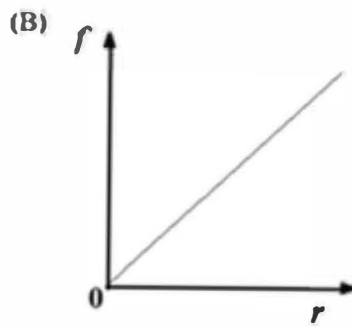
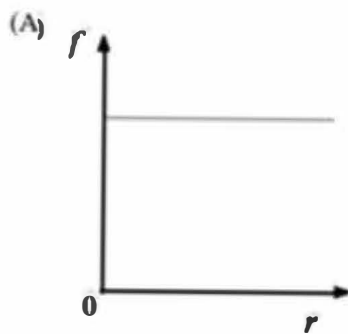
- (A) Still 13.7 billion years
 (B) 3.43 billion years
 (C) 6.85 billion years

(D) 27.4 billion years

14. A negatively charged particle in a uniform magnetic field B moves in a circular path of radius r , as shown.



Which of the following graphs best depicts how the frequency of revolution, f , of the particle depends on the radius r ?



15. A person aims an arrow at a target 15 m away, with the bullseye at the same height as the arrow's release point.

If the arrow leaves the bow with a speed of 20 m/s, what angle above the horizontal must the arrow be launched to hit the bullseye if air resistance is negligible?

- (A) 1.0°
 (B) 10.8°
 (C) 15.2°
 (D) 30.3°

16. In Millikan's experiment, the charge on each drop of oil was measured in coulombs. Imagine the same experiment, but with charges measured in a fictitious unit called a halo (H).

The results are shown in the table to the right.

What is the charge on an electron in halos?

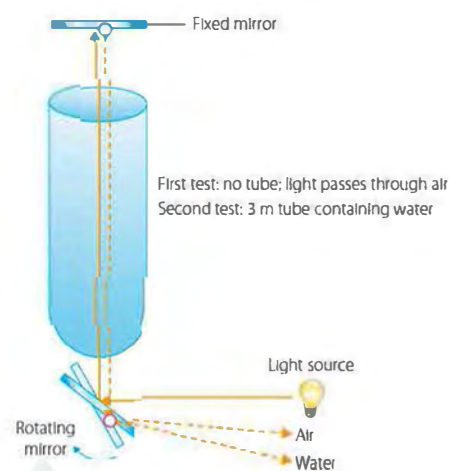
- (A) 2.6×10^{-15}
 (B) 6.0×10^{-15}
 (C) 3.6×10^{-14}
 (D) 9.6×10^{-14}

Drop	Charge (H)
A	3.6×10^{-14}
B	6.0×10^{-14}
C	6.6×10^{-14}
D	7.2×10^{-14}
E	9.6×10^{-14}

17. The following diagram shows Foucault's experiment to measure the speed of light. A narrow beam of light is directed at a rotating mirror. The light travels 9.0 m to a fixed mirror, reflects back, and returns to the rotating mirror. Because the mirror has rotated slightly during the time it takes the light to travel to the fixed mirror and back, the returning light beam is deflected and does not return to the exact point of origin.

The total round-trip time of the light in air is measured to be 60 nanoseconds.

A 3.0 m long tube filled with water is then placed along the light path between the mirrors. The refractive index of water is $n = 1.33$.



What is the expected time delay (in nanoseconds) when the light travels 3.0 m through water instead of air?

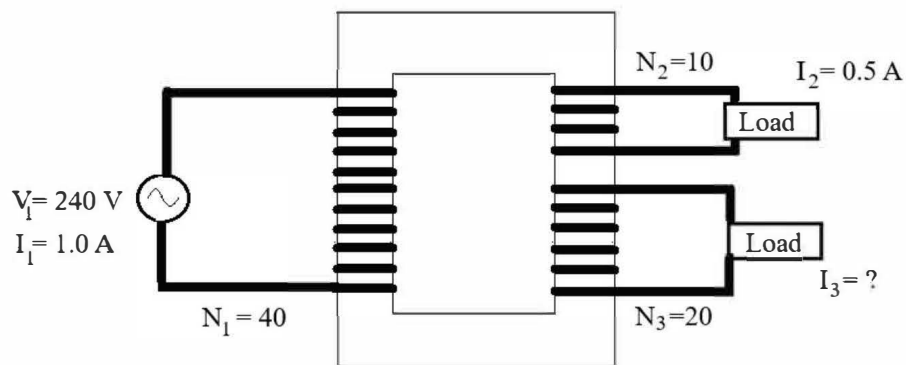
- (A) 1.33
 (B) 2.26
 (C) 3.27
 (D) 13.27

18. A spaceship flies past Earth at a speed of $0.8c$. Two observers on Earth are spaced 56.5m apart.

How far apart, in metres, does the spaceship crew measure the observers sitting apart?

- (A) 25.3
 (B) 33.9
 (C) 45.2
 (D) 50.5

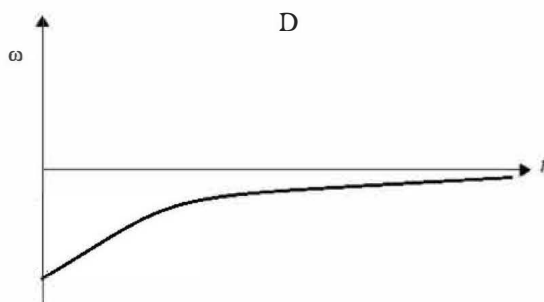
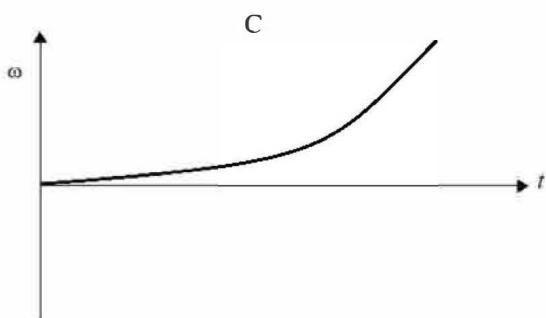
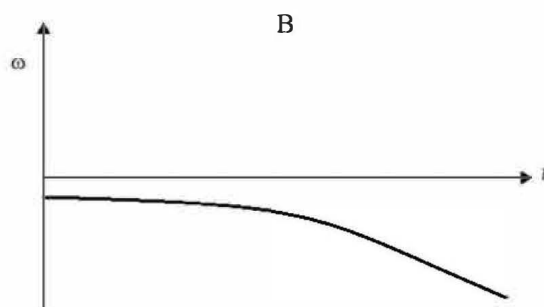
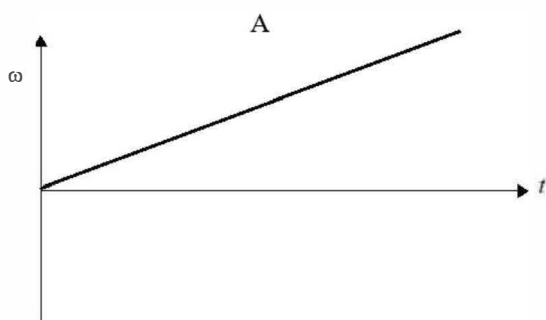
19. A transformer has a single primary coil of 40 turns with a voltage of 240 V and a current of 1 A. There are two secondary coils, each with 10 and 20 turns. The secondary coil with 10 turns has a current of 0.5 A, as shown below.



What is the current, in A, in the 20-turn secondary coil?

- (A) 0.875
- (B) 1.50
- (C) 1.75
- (D) 3.00

20. The graph shown below shows the angular velocity of a rotating wheel.



Which graph shows a wheel slowing down?

- (A) A
- (B) B
- (C) C
- (D) D

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Student Number

2025

Parramatta High School

HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Physics

Section I Part A - Multiple Choice

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$

(A) 2

(B) 6

(C) 8

(D) 9

A ☐

B ☒

C ☐

D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒

B ☒

C ☐

D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒

B ☒

C ☐

D ☐

Correct

Start Here	1	AO	BO	CO	DO	11	AO	BO	CO	DO
	2	AO	BO	CO	DO	12	AO	BO	CO	DO
	3	AO	BO	CO	DO	13	AO	BO	CO	DO
	4	AO	BO	CO	DO	14	AO	BO	CO	DO
	5	AO	BO	CO	DO	15	AO	BO	CO	DO
	6	AO	BO	CO	DO	16	AO	BO	CO	DO
	7	AO	BO	CO	DO	17	AO	BO	CO	DO
	8	AO	BO	CO	DO	18	AO	BO	CO	DO
	9	AO	BO	CO	DO	19	AO	BO	CO	DO
	10	AO	BO	CO	DO	20	AO	BO	CO	DO

Section II

80 marks

Attempt Questions 21-32

Allow about 2 hours and 25 minutes for this section.

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

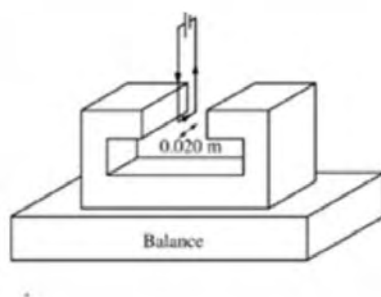
Show all relevant working out in questions involving calculations.

Extra writing space is provided in an extra writing booklet. If you use this space, clearly indicate which question you are answering.

Question 21 (8 marks)

A rectangular wire loop, 0.72 m in length, is used in an experiment to measure the strength of the magnetic field between the poles of a magnet. The magnet is placed on a digital balance, and the wire loop is held fixed between the poles of the magnet. The 0.020 m long horizontal segment of the loop is midway between the poles and perpendicular to the direction of the magnetic field.

The power supply in the loop is turned on, allowing a 4.0 A current to flow in the direction shown.



- a) The reading on the balance increased by 0.060 N when the power supply was turned on.

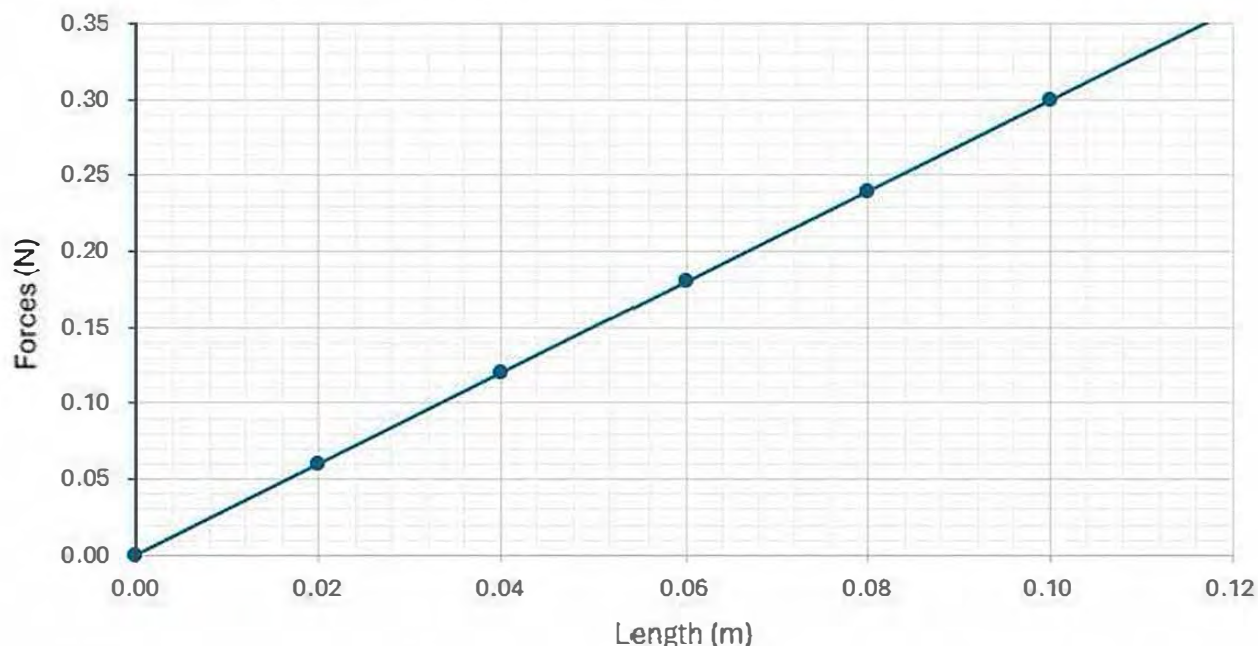
Determine the magnetic field strength. Justify your answer (3)

Question 21 continues on page 12

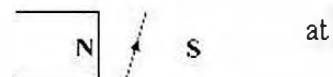
Question 21 continued

- b) Various rectangular loops with the same total length of wire were constructed such that the lengths of the horizontal segments of the wire loops varied between 0.02 m and 0.10 m. The horizontal segment of each loop was always centred between the poles, and the current in each loop was always 4.0 A.

The following graph illustrates the theoretical relationship between the magnitude of the force on the magnet and the length of the wire.



- i. Suppose the wire segments were misaligned and placed a constant non-perpendicular angle to the magnetic field set up as shown below.

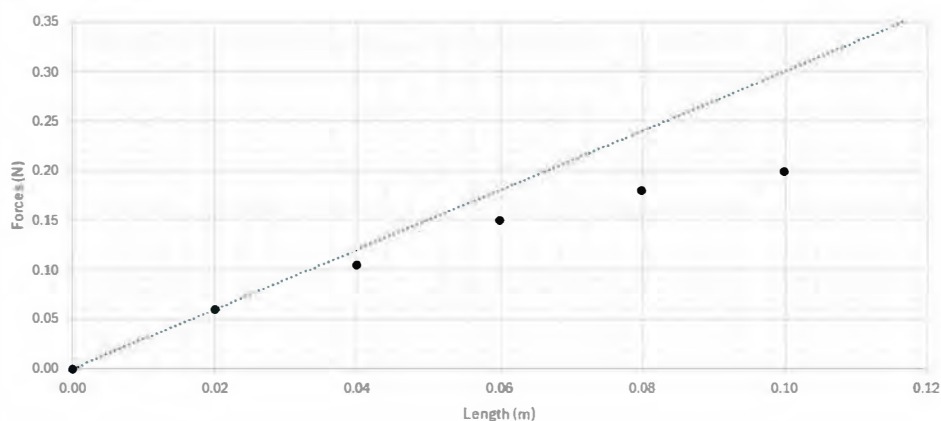


On the graph above, sketch a possible relationship between the magnitude of the force on the magnet and the length of the wire segment. Justify your answer. (3)

Question 21 continues on page 13

Question 21 continued

- ii. Suppose the loops are correctly placed perpendicular to the field, and the data below is obtained.



Explain a likely cause of the discrepancy between the data and the theoretical relationship.

(2)

Question 22 (8 marks)

The nucleus plays a key role in nuclear reactions.

- a) A sample of radioactive isotope is found to have an activity of 145 Bq immediately after it is pulled from the reactor that forms the isotope. Its activity, 1h and 35 minutes later, is measured to be 92.1 Bq.

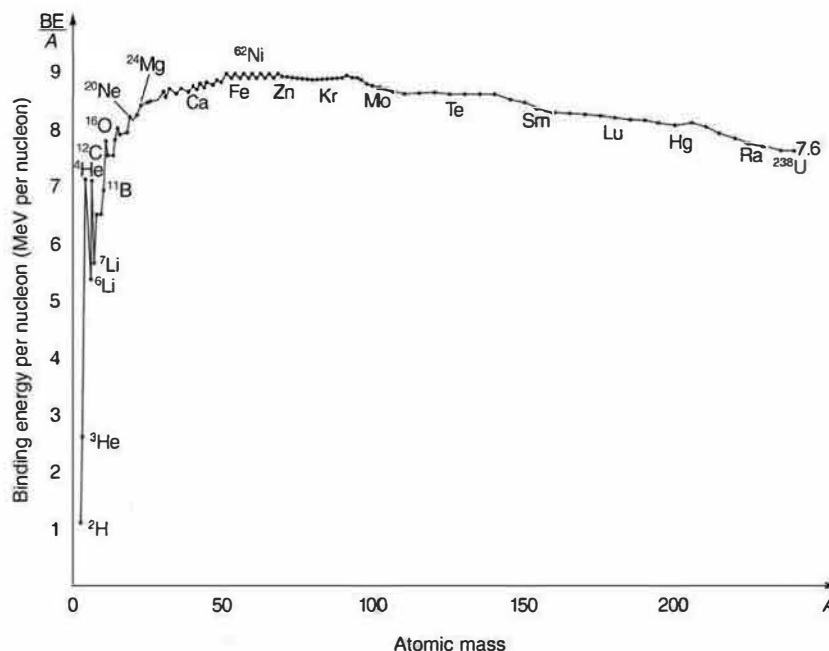
Find the half-life of the sample in hours.

(3)

Question 22 continues on page 14

Question 22 continued

b) The graph shows the relationship between binding energy and atomic mass.



- i) Explain whether it would require more energy to take apart a ^{235}U nucleus than it would to take apart a ^{16}O nucleus. (2)

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- ii) With reference to the graph, explain why in the very early universe, shortly after the Big Bang, a large fraction of the free nucleons formed into ^4He , so much so that other elements were extremely rare. (3)

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End of Question 22

Question 23 (5 marks)

A beam of particles, Y, of charge $q = +3.2 \times 10^{-19} \text{ C}$ and mass $m = 6.68 \times 10^{-26} \text{ kg}$ enters region I with a range of velocities all in the direction shown in the diagram below.



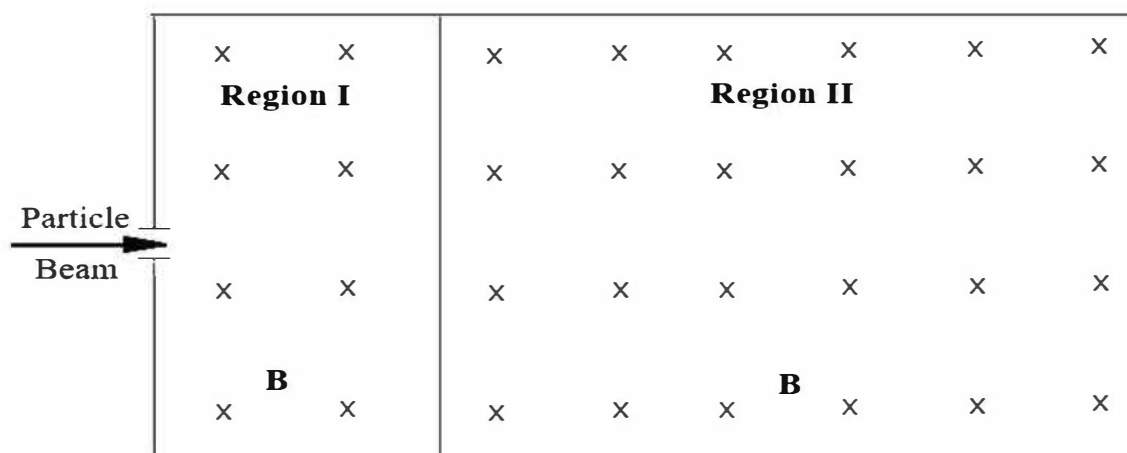
There is a magnetic field in Region I directed into the page with magnitude $B = 0.12 \text{ T}$. Charged metal plates are placed in appropriate locations to create a uniform electric field of magnitude $E = 5000 \text{ N/C}$ in Region I. As a result, some of the charged particles pass straight through Region I undeflected. Gravitational effects are negligible.

The particles that pass straight through enter Region II, in which there is no electric field, and the magnetic field has the same magnitude and direction as in Region I. The path of the particles in Region II is a circular arc of radius R .

- a) Calculate the radius R . (3)

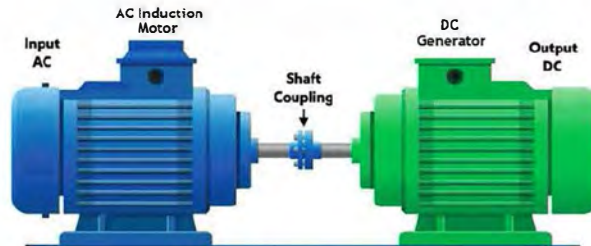
- b) The beam of particles Y is replaced with a beam of particles Z, which has a charge $q = -3.2 \times 10^{-19} \text{ C}$ and half the mass of particle Y.

On the following diagram, sketch the resulting paths of the two particles. (2)



Question 24 (7 marks)

The picture below shows a motor-generator set, a device that converts electrical energy into another form—typically from AC to DC—by mechanically coupling a motor and a generator on the same shaft.



<https://www.electricaltechnology.org/2022/01/motor-generator-set.html>

- a) Compare the components of the motor to the generators. (3)

- b) How can you improve the efficiency of the motor-generator set? (4)

[illegible]

Question 25 (9 marks)

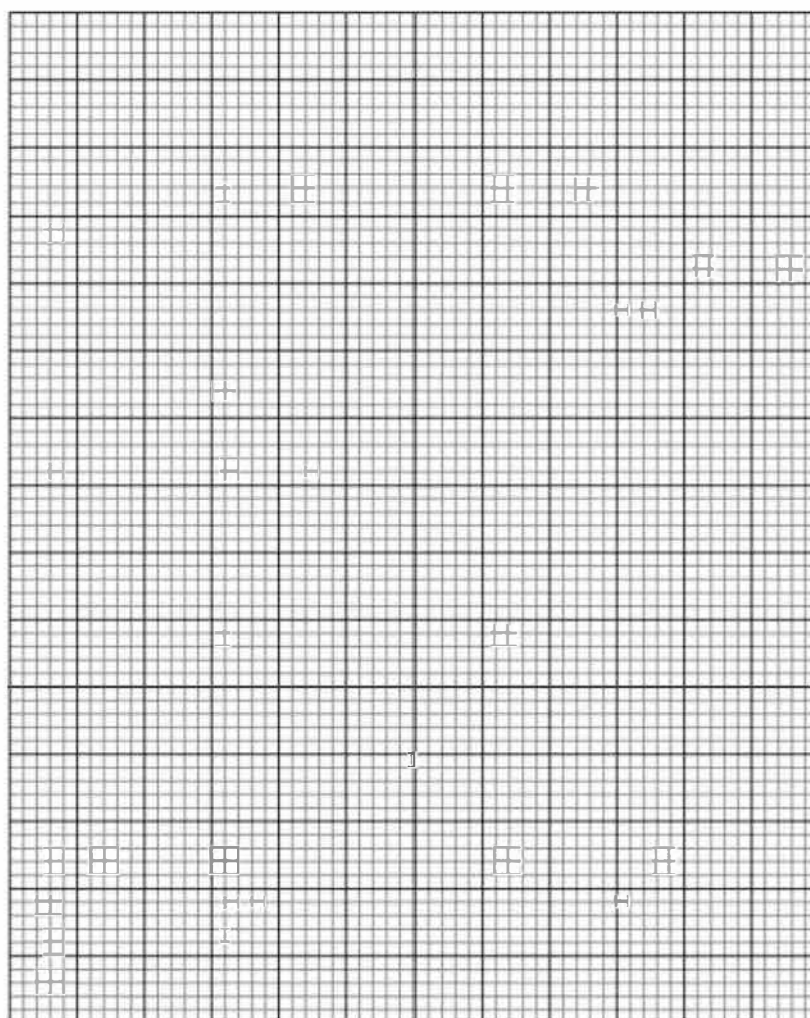
A student performed the photoelectric effect experiment on a sodium metal plate. The stopping voltage for the experiment was 0.39 V.

The following table shows their results.

<i>Kinetic energy of photoelectrons (10^{-19} J)</i>	<i>Frequency ($\times 10^{14} \text{ Hz}$)</i>
1.46	5.64
2.76	7.59
4.35	10.00
7.33	14.49
11.00	20.00

a) Draw a graph for the results

(3)



Question 25 continues on page 18

Question 25 continued

b) Explain how the student can measure the stopping voltage.

(2)

[illegible]

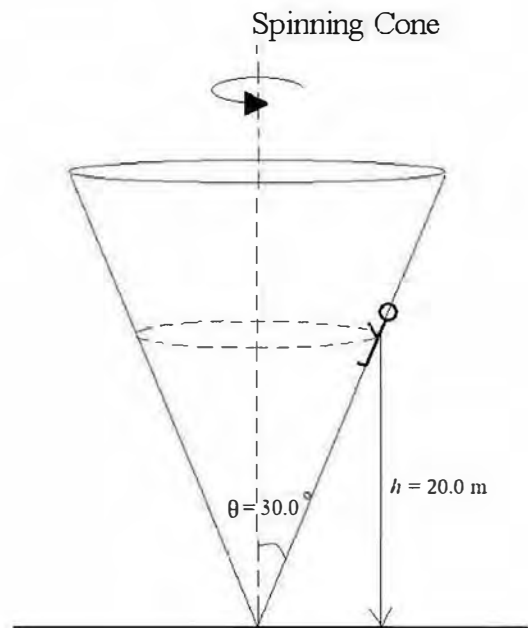
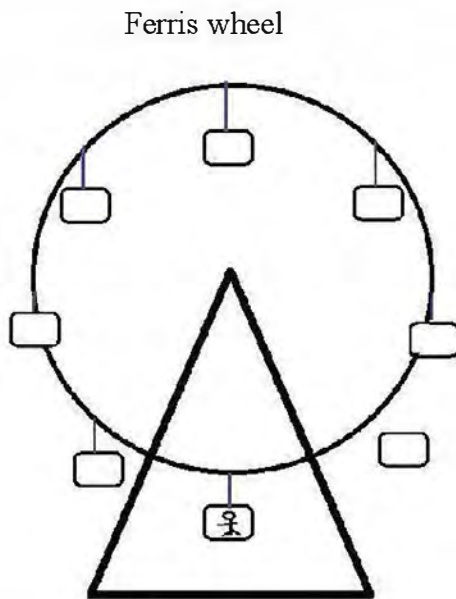
c) Assess the accuracy of the student's results.

(4)

End of Question 25

Question 26 (9 marks)

The picture shows two rides at a theme park, the “Ferris wheel” and the “Spinning Cone”. In motion, both rides move at constant speed.



- a) Discuss the total energy and work done on a passenger in the Ferris wheel and the Spinning cone. In your answer, discuss how gravitational potential energy, kinetic energy, and work done by forces change during the motion. (4)

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Question 26 continues on page 20

Question 26 continued

- b) During the ride on the spinning cone, the cone-shaped enclosure spins about a vertical axis, causing the passenger to rotate in a horizontal circle at a height of 20.0 metres. The coefficient of static friction between the wall and the passenger is 0.650.

Determine the minimum and maximum speeds required so that the passenger does not slip down or up the wall, respectively. (5)

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End of Question 26

Question 27 (4 marks)

Explain the relationship between the supplied emf, back emf, and net emf in a motor as it starts up and continues to spin. Support your answer with a labelled graph. (4)

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Question 28 (4 marks)

Discuss the impacts Einstein's theory of Special Relativity will have on space travel. (4)

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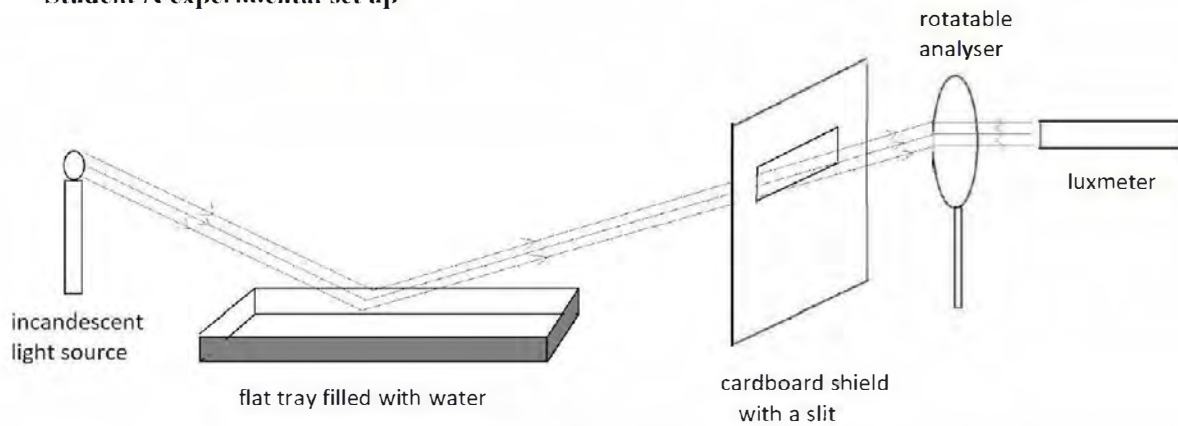
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Question 29 (6 marks)

The following diagram shows an experiment that investigates the nature of light.

Student A experimental set up



An initial reading of 2,000 lux was recorded. However, when the analyser was rotated 90° , the luxmeter recorded a reading of 7 lux.

Explain how the experiment supports Maxwell's theory of electromagnetism.

(6)

(The page contains faint horizontal lines suggesting bleed-through from the reverse side.)

Question 30 (7 marks)

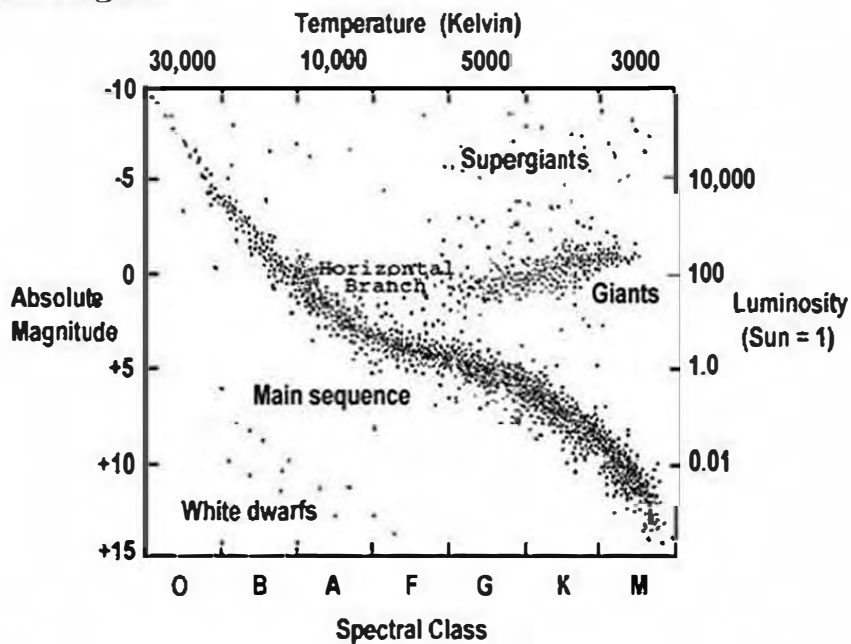
Explain how the study of circular motion and waves helped scientists develop and change the models of the atom over time. (7)

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

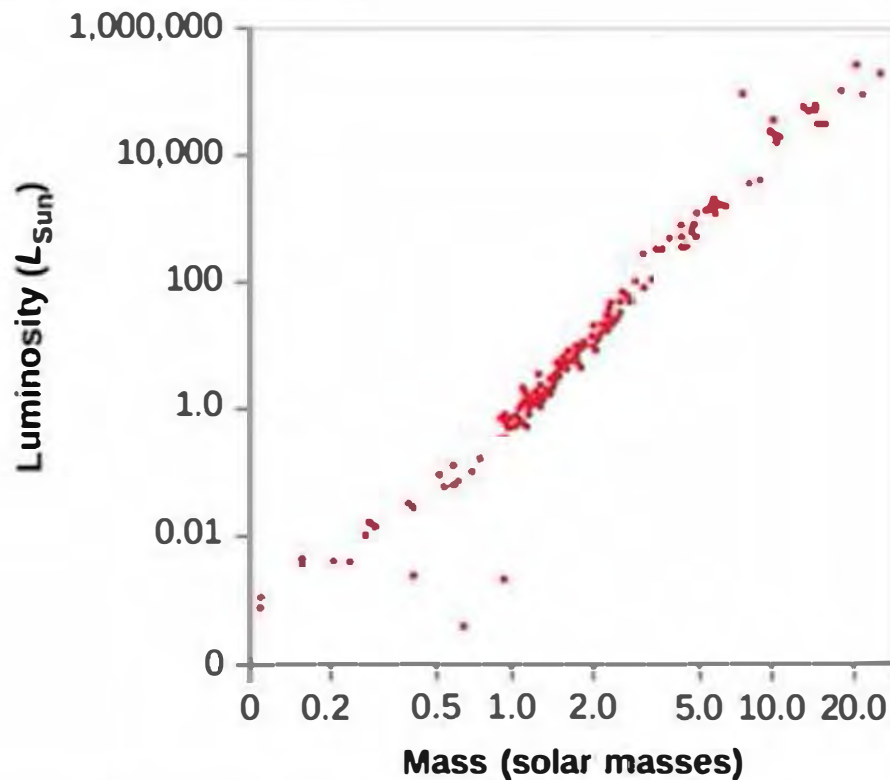
Question 31 (8 marks)

Scientists frequently utilise graphical relationships to model systems, explain underlying mechanisms, and make predictions about natural phenomena. For example, by interpreting stellar spectra, we know our Sun is a G star with a surface temperature of 5,772 K.

Graph 1- H-R Diagram



Graph 2- Luminosity Vs Mass

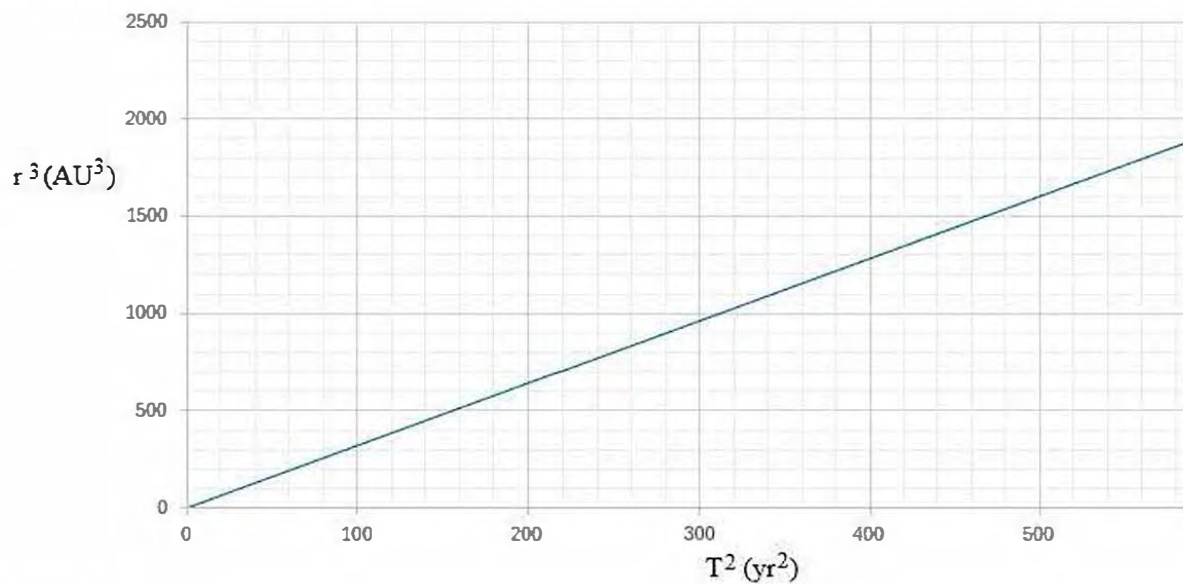


Mass of the Sun = 1.989×10^{30} kg

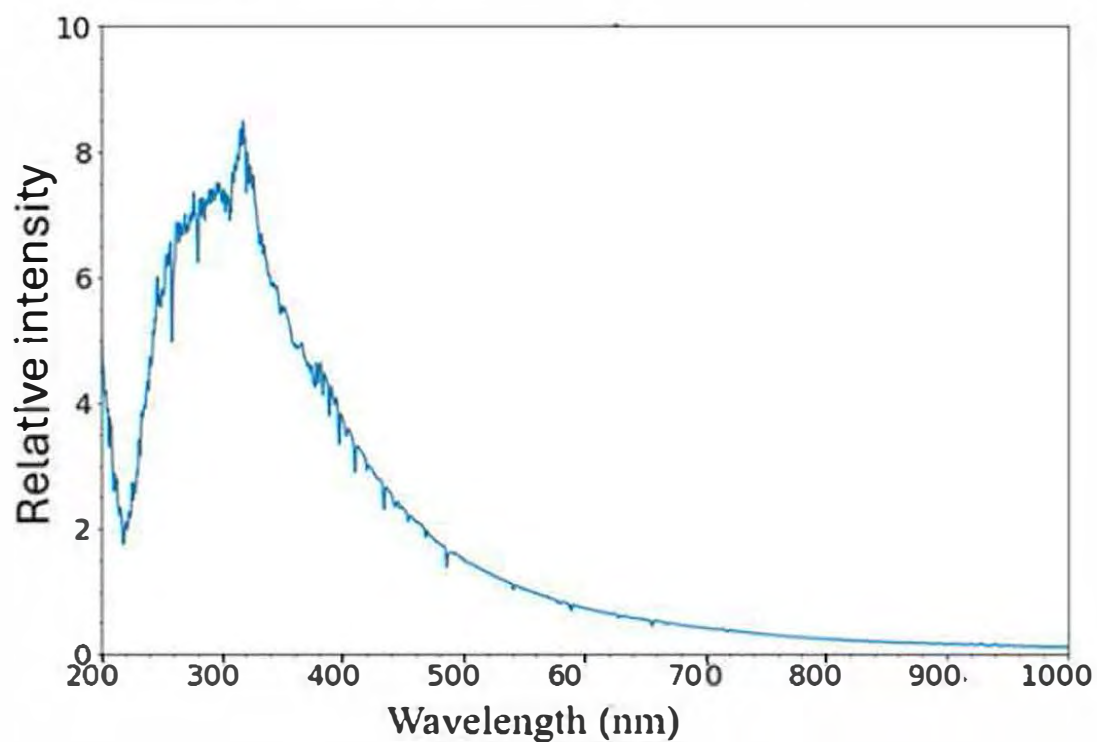
Question 31 continues on page 26

Question 31 continued

Graph 3 – R^3 vs T^2 for satellites orbiting Star X



Graph 4- Relative intensity vs wavelength for Star X



Question 31 continues on page 27

Question 31 continued

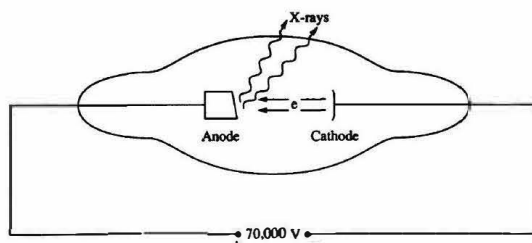
By interpreting these graphical relationships in combination, describe how the physical and chemical properties of Star X differ from those of our Sun, including their chemical processes. (8)

[The page contains faint horizontal lines indicating where writing should occur.]

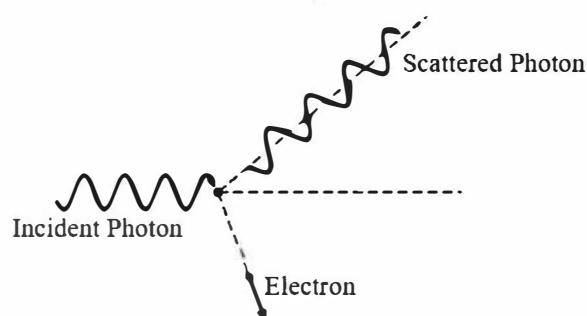
End of Question 31

Question 32 (5 marks)

In the X-ray tube shown above, a potential difference of **70,000** volts is applied across the two electrodes. Electrons emitted from the cathode are accelerated to the anode, where X-rays are produced.



An X-ray photon of the maximum energy produced by this tube leaves the tube and collides elastically with an electron at rest. As a result, the electron recoils and the X-ray is scattered, as shown above. The frequency of the scattered X-ray photon is 1.64×10^{19} hertz.



Determine the de Broglie wavelength of the electron.

(5)

This image shows a full page of a document template designed for handwriting practice. It consists of ten evenly spaced, horizontal dashed lines extending across the entire width of the page. The background is plain white, providing a clear guide for letter height and placement. There are no margins, text, or other markings present.

End of Test

Blank Page

ANSWERS

Start
Here

1	AO	BO	CO	DO	11	AO	BO	CO	DO
2	AO	BO	CO	DO	12	AO	BO	CO	DO
3	AO	BO	CO	DO	13	AO	BO	CO	DO
4	AO	BO	CO	DO	14	AO	BO	CO	DO
5	AO	BO	CO	DO	15	AO	BO	CO	DO
6	AO	BO	CO	DO	16	AO	BO	CO	DO
7	AO	BO	CO	DO	17	AO	BO	CO	DO
8	AO	BO	CO	DO	18	AO	BO	CO	DO
9	AO	BO	CO	DO	19	AO	BO	CO	DO
10	AO	BO	CO	DO	20	AO	BO	CO	DO

19

$$V_s = \frac{10+20}{40} \times 240 = 180V$$

The average power for the primary coil = $240 \times 1 = 240W$

For the secondary coil with 10 turns, $V=60V$ $P_{ave} = 60 \times 0.5 = 30W$

$P_{in} = P_{out}$

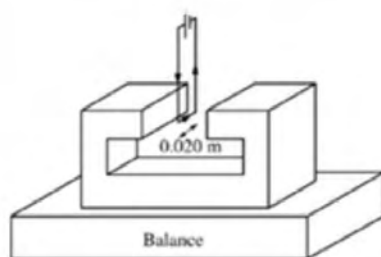
So in 20 turns, secondary coil with 120V, $P = 240W - 30W = 210W$

$$I = \frac{210}{120} = 1.75 (C)$$

Question 21 (8 marks)

A rectangular wire loop, 0.72 m in length, is used in an experiment to measure the strength of the magnetic field between the poles of a magnet. The magnet is placed on a digital balance, and the wire loop is held fixed between the poles of the magnet. The 0.020 m long horizontal segment of the loop is midway between the poles and perpendicular to the direction of the magnetic field.

The power supply in the loop is turned on, allowing a 4.0 A current to flow in the direction shown.



Not to scale

- a) The reading on the balance increased by 0.060 N when the power supply was turned on.

Determine the magnetic field strength. Justify your answer

(3)

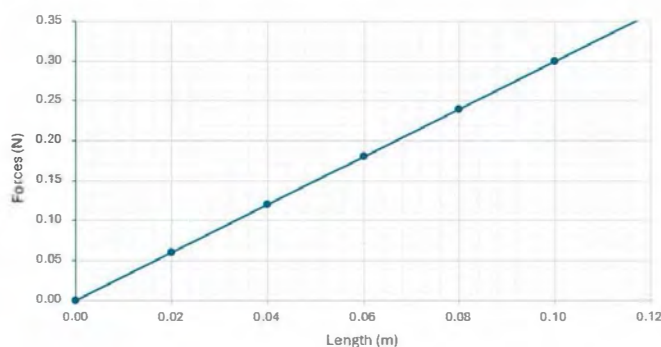
Marking Criteria	Marks
• Calculates the magnetic field with a complete justification	3
• Calculates the magnetic field with justification	2
• Provide a relevant step	1

Sample answer

$F_b = BIL$ $0.06 \text{ N} = B(4)(0.02)$ $B = 0.75 \text{ T}$ from left to right, since balance when heavier, therefore than the force on the magnet will be down, thus the force on the wire is up based on Newton's 3rd law of motion, which means the magnetic field is from the right to left.

- b) Various rectangular loops with the same total length of wire were constructed such that the lengths of the horizontal segments of the wire loops varied between 0.02 m and 0.10 m. The horizontal segment of each loop was always centred between the poles, and the current in each loop was always 4.0 A.

The following graph illustrates the theoretical relationship between the magnitude of the force on the magnet and the length of the wire.



- i. Suppose the wire segments were misaligned and placed a constant non-perpendicular angle to the magnetic field set up as shown below.

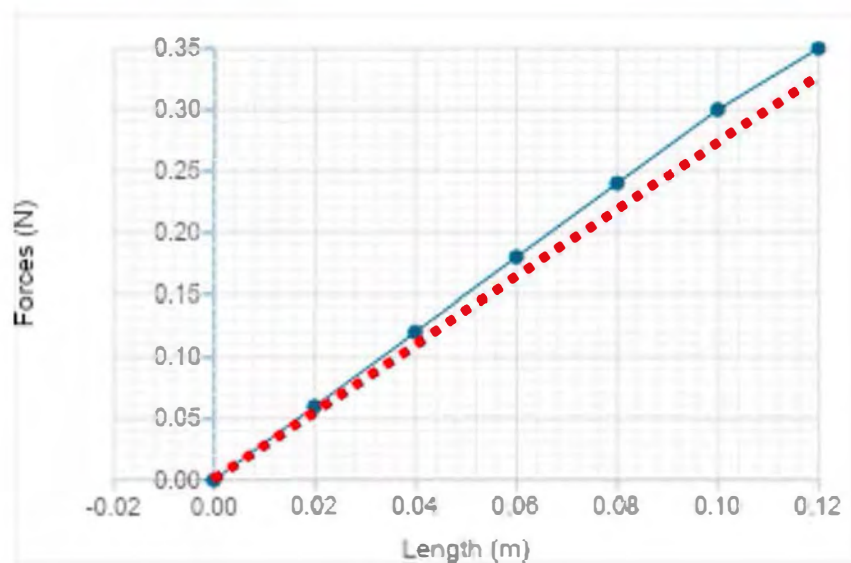


On the graph above, sketch a possible relationship between the magnitude of the force on the magnet and the length of the wire segment. Justify your answer. (3)

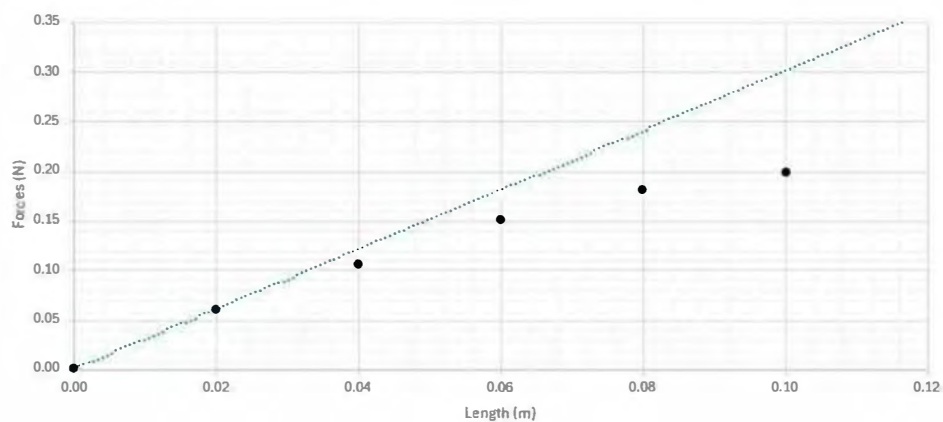
Marking Criteria	Marks
• Sketch the relationship with complete reasoning	3
• Sketch the relationship with some reasoning	2
• Provide a relevant information	1

Sample answer

Since the length of the wire is not perpendicular, only a component of the B field is used to determine the force. So in the equation $F_b = BIL$, the B value is reduced, and for increasing lengths, there should be less and less force compared to the ideal line shown.



- ii. Suppose the loops are correctly placed perpendicular to the field, and the data below is obtained.



Explain a likely cause of the discrepancy between the data and the theoretical relationship.

(2)

Marking Criteria	Marks
• Provides an explanation for a likely cause	2
• Identify a cause	1

Sample answer

The graph shows a decreasing magnetic force, but it's not due to the wire's misalignment.

One explanation could be that as the wire segment in between the magnets got longer, it moved outward away from the poles of the N-S magnet, and some of the wire had a smaller field acting on it compared to the parts in the centre of the magnet.

OR

The graph shows a decreasing magnetic force, but it's not due to the wire's misalignment.

The source of error is that as the bottom part of the loop gets longer, the sides become shorter, bringing the top part of the loop lower, and the top part of the loop will begin to exert a force in the opposite direction, thereby lessening the net force on the wire.

Question 22 (8 marks)

The nucleus plays a key role in nuclear reactions.

- a) A sample of radioactive isotope is found to have an activity of 145 Bq immediately after it is pulled from the reactor that forms the isotope. Its activity, 1h and 35 minutes later, is measured to be 92.1 Bq.

Find the half-life of the sample in hours.

(3)

Marking Criteria	Marks
• Calculates the half-life in hrs	3
• Provide steps to calculate half-life	2
• Provide a relevant step	1

Sample answer

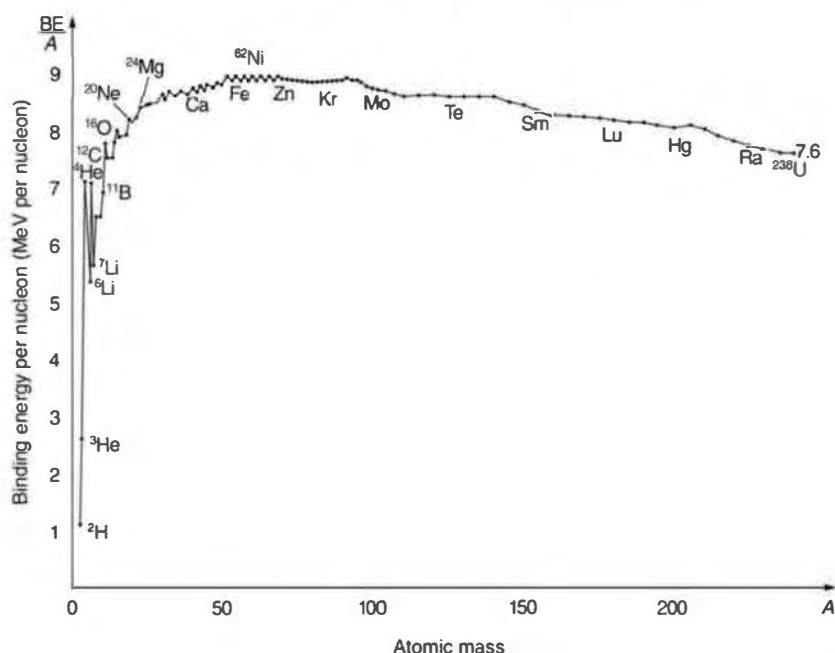
$$N_0 = 145 \text{ Bq}, N_t = 92.1 \text{ Bq}, t = 1 + \frac{35}{60} = \frac{19}{12}$$

$$N_t = N_0 e^{-\lambda t}, \lambda = \frac{\ln(\frac{92}{145})}{\frac{19}{12}} = \frac{\ln(0.6344827586)}{1.58333333333}$$

$$t_{\frac{1}{2}} = \frac{\ln(2)}{\lambda} = \frac{\ln(2)}{\frac{\ln(\frac{92}{145})}{\frac{19}{12}}} = 2.41234135 = 2.41 \text{ hr}$$

Question 22 continued

b) The graph shows the relationship between binding energy and atomic mass.



- i) Explain whether it would require more energy to take apart a ^{235}U nucleus than it would to take apart a ^{16}O nucleus. (2)

Marking Criteria	Marks
Comparing the value of the binding energy for ^{16}O and ^{235}U in the graph to explain why ^{16}O requires more energy.	2
Provide some relevant information	1

Sample answer

From the graph, the binding energy for ^{235}U is 7.6, whereas the binding energy for ^{16}O is greater than 7.6, therefore, it takes more energy to split the nucleus of ^{16}O apart.

- ii) With reference to the graph, explain why in the very early universe, shortly after the Big Bang, a large fraction of the free nucleons formed into ^4He , so much so that other elements were extremely rare. (3)

Marking Criteria	Marks
Provide a thorough explanation with reference to the graph	3
Provide a sound explanation with reference to the graph	2
Provide an explanation	1

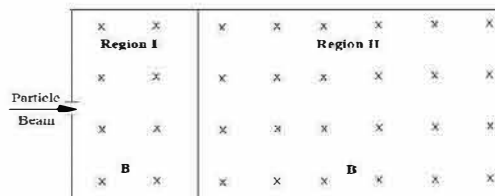
Sample answer

Shortly after the Big Bang, temperatures and densities were extremely high, allowing nuclear fusion to occur. According to the binding energy per nucleon graph, nuclei with higher binding energy per nucleon are more stable. Among light nuclei, helium-4 (${}^4\text{He}$) has a particularly high binding energy per nucleon 7.2, meaning it is very stable compared to other light elements like deuterium 1.2, tritium 2.7, or lithium 5.2.

Because ${}^4\text{He}$ is so stable, once free protons and neutrons (nucleons) began to combine, they quickly formed deuterium and then fused further into ${}^4\text{He}$. This process released energy and was energetically favourable. As a result, a large fraction of the available nucleons ended up locked in ${}^4\text{He}$ nuclei. There was not enough time or favourable conditions for significant amounts of heavier elements to form before the universe cooled and fusion reactions stopped.

Question 23 (5 marks)

A beam of particles, Y, of charge $q = +3.2 \times 10^{-19} \text{ C}$ and mass $m = 6.68 \times 10^{-26} \text{ kg}$ enters region I with a range of velocities all in the direction shown in the diagram below.



There is a magnetic field in Region I directed into the page with magnitude $B = 0.12 \text{ T}$. Charged metal plates are placed in appropriate locations to create a uniform electric field of magnitude $E = 5000 \text{ N/C}$ in Region I. As a result, some of the charged particles pass straight through Region I undeflected. Gravitational effects are negligible.

The particles that pass straight through enter Region II, in which there is no electric field, and the magnetic field has the same magnitude and direction as in Region I. The path of the particles in Region II is a circular arc of radius R .

a) Calculate the radius R .

(3)

Marking Criteria	Marks
• Calculates the radius with sig fig	3
• Provide steps to calculate radius	2
• Provide a relevant step	1

$$F_e = F_b \text{ therefore } Eq = qvB \text{ so } v = E/B = 5000/0.12 = 41666.6667$$

$$\text{Using } F_{\text{net}(c)} = mv^2/r$$

$$qvB = mv^2/r$$

$$r = mv / Bq$$

$$r = (6.68 \times 10^{-26})(41666.6667) / (0.12)(3.2 \times 10^{-19}) = 0.072 \text{ m}$$

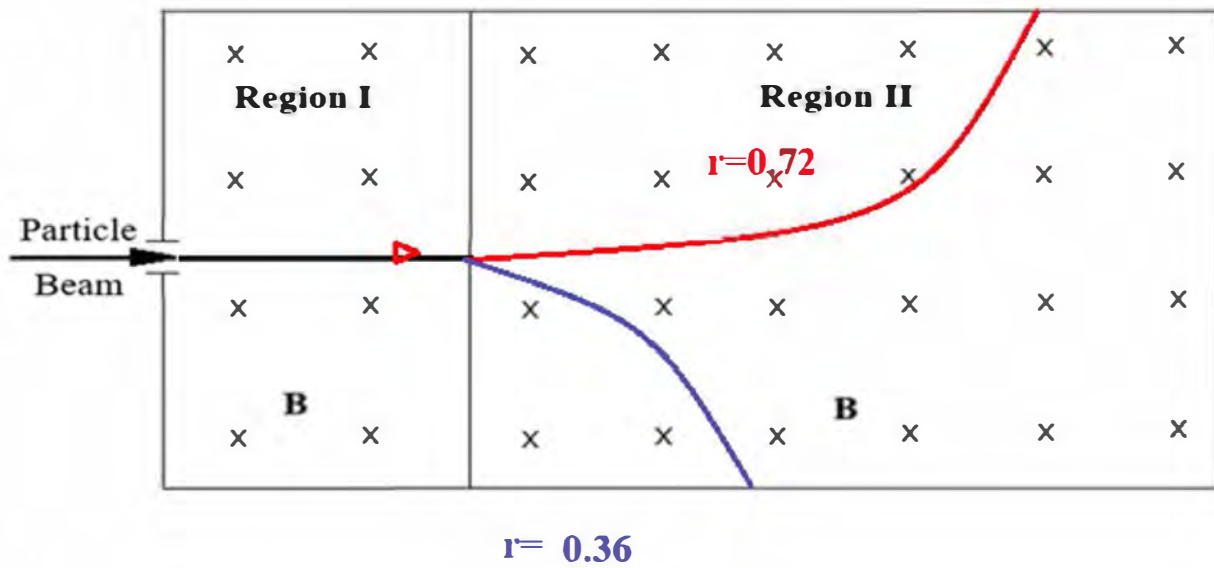
b) The beam of particles Y is replaced with a beam of particles Z, which has a charge $q = -3.2 \times 10^{-19} \text{ C}$ and half the mass of particle Y.

On the following diagram, sketch the resulting paths of the two particles.

(2)

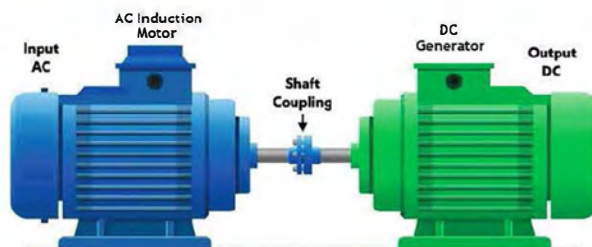
Marking Criteria	Marks
• Draws the resulting paths of the two particles.	2
• Draw the resulting path for one particle.	1

Sample answer



Question 24 (7 marks)

The picture below shows a motor-generator set, a device that converts electrical energy into another form—typically from AC to DC—by mechanically coupling a motor and a generator on the same shaft.



<https://www.electricaltechnology.org/2022/01/motor-generator-set.html>

c) Compare the components of the motor to the generators. (3)

Marking Criteria	Marks
• Compare the components of the motor and the generators	3
• List the components of the motor and/or the generators	2
• Identify a component of a motor or generator	1

Sample answer

An AC induction motor and a DC generator both have a stator and a rotor, but their components serve different functions. In an AC induction motor, the stator is supplied with AC to create a rotating magnetic field, which induces current in the rotor to produce torque, and no commutator or brushes are needed. In contrast, a DC generator has a stator that provides a magnetic field and a rotating armature where EMF is induced; it uses a commutator and brushes to convert this into a steady DC output.

d) How can you improve the efficiency of the motor-generator set? (4)

Marking Criteria	Marks
• Provides a thorough explanation	4
• Provides a sound explanation	3
• Outlines ways	2
• Provide some relevant information	1

Sample answer

For the AC induction motor, use higher-quality materials, such as copper windings, instead of aluminium for lower resistance and better conductivity. Using high-grade electrical steel for the stator core reduces hysteresis and eddy current. Using thin, laminated cores reduces eddy current.

For the DC generator, use thicker or high-conductivity copper windings to reduce electrical resistance in the armature and field coils. For the iron core, use high-grade silicon steel laminations to reduce hysteresis losses and eddy currents. Use low-friction bearings and lubricate them properly to reduce frictional drag. For the brushes, use carbon or graphite brushes suited for the specific current and speed range, maintaining good contact pressure. Clean the commutator to reduce contact resistance and energy loss through arcing.

These all reduce heat loss to improve efficiency.

Question 25 (9 marks)

A student performed the photoelectric effect experiment on a sodium metal plate. The stopping voltage for the experiment was 0.39 V.

The following table shows their results.

<i>Kinetic energy of photoelectrons ($10^{-19} J$)</i>	<i>Frequency ($\times 10^{14} Hz$)</i>
1.46	5.64
2.76	7.59
4.35	10.00
7.33	14.49
11.00	20.00

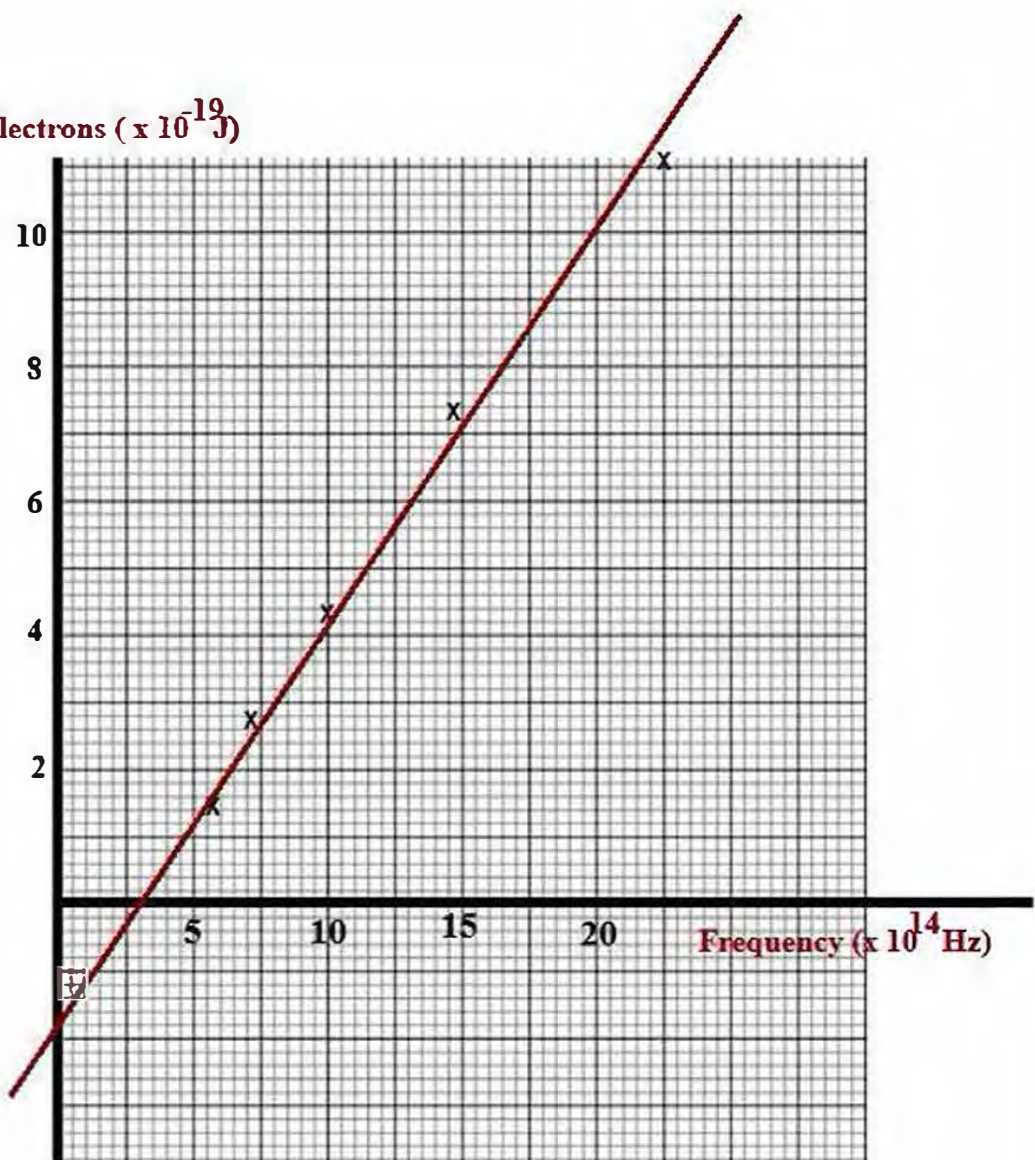
a) Draw a graph for the results

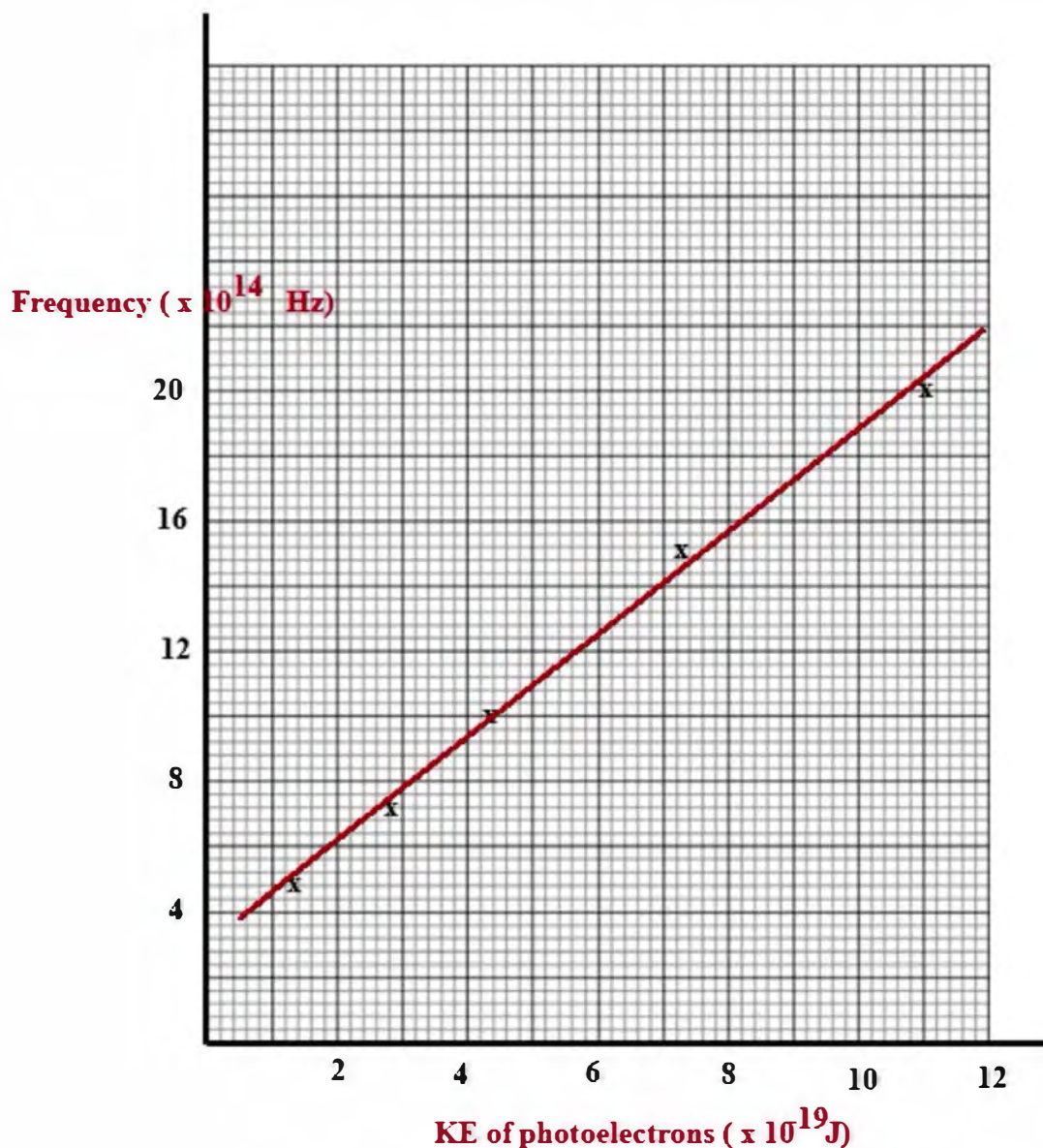
(3)

Marking Criteria	Marks
• Completely labelled graph with correct data plot and trendline	3
• Completely labelled graph with correct data plot or trendline	2
• Provide some relevant information	1

Sample answer

KE of photoelectrons ($\times 10^{-19}$ J)





b) Explain how the student can measure the stopping voltage. (2)

Mark	Criteria
3	Explains (what is done and why it is done) HOW stopping voltage is measured
2	Outlines (what is done without reasoning/why)
1	Outlines a step OR definition of stopping voltage

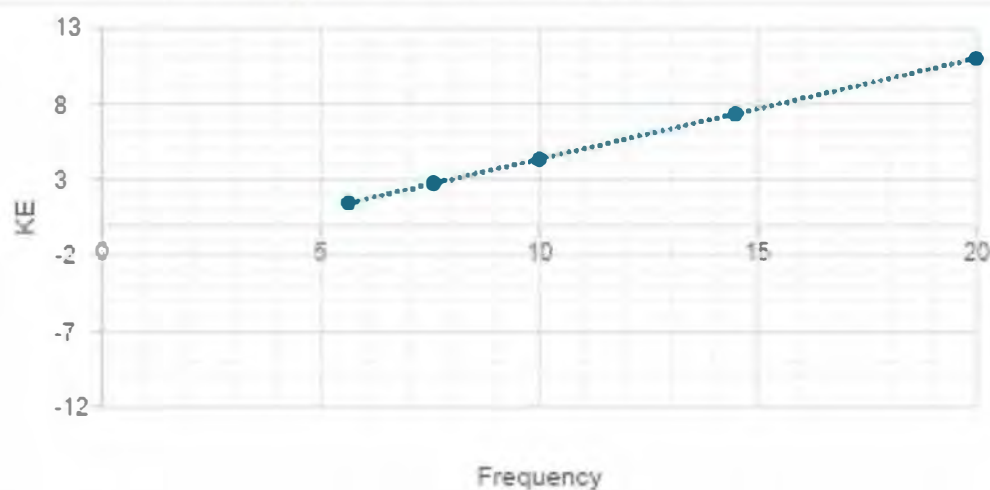
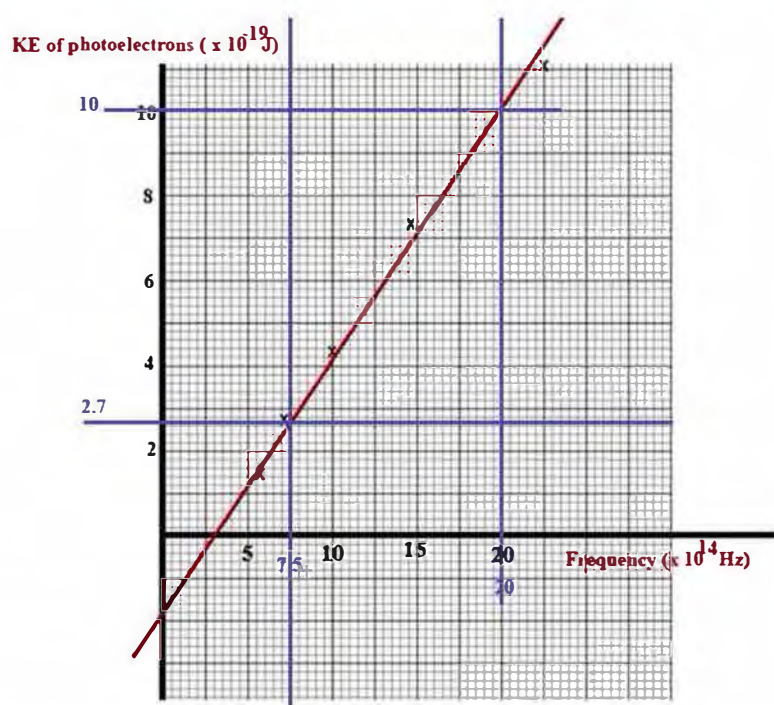
Stopping voltage is the energy required to stop the electrons ejected from photoelectric effect to reach the positive terminal. To do this, apply a potential difference that reverses the cathode and anode of the CRT. This will apply a force that will hinder the ejected electrons from the sodium metal. Place a voltmeter in the circuit to measure the voltage applied. When the potential difference

is large enough to stop all ejected electrons from reaching the opposite terminal, this is the stopping voltage. The maximum kinetic energy of the electrons can be calculated from this: $KE_{\text{max}} = eV$

c) Assess the accuracy of the student's results.

(4)

Marking Criteria	Mark
Assess the accuracy of the student's result by using the graph to calculate the gradient. Calculate the percentage error and make a judgment.	4
Assess the accuracy of the student's result by using the graph to calculate the gradient. Compare the result with Planck's constant and make a judgment.	3
Using a point to calculate the gradient. Compare the result with Planck's constant and make a judgment.	2
Calculated the gradient correctly OR Outlined how the accuracy has to do with how close the gradient is to Plank's constant.	1

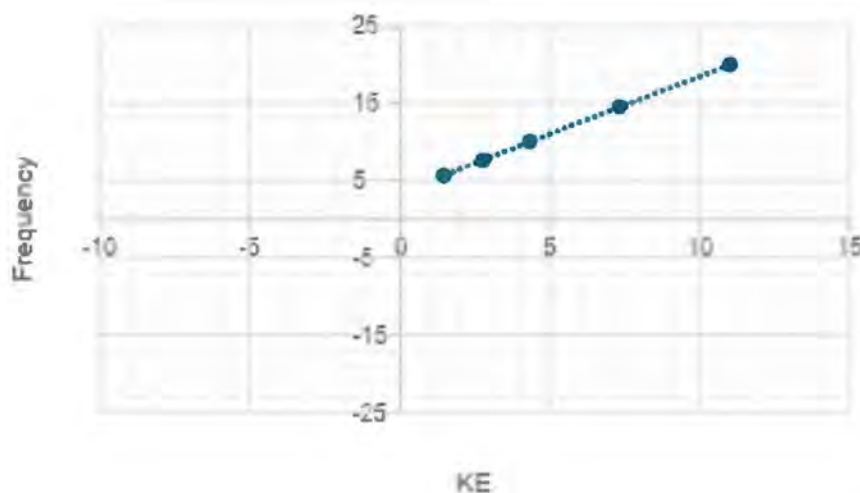


$$E = hf - \phi$$

At $\phi = 0$, $E = hf$, therefore $h = \text{gradient}$

$$\text{gradient} = 0.6641 \times \frac{10^{-19}}{10^{14}} = 6.641 \times 10^{-34}$$

$$\text{Percentage error} = \frac{6.641 \times 10^{-34} - 6.626 \times 10^{-34}}{6.626 \times 10^{-34}} = 0.23\% \text{ so it is very accurate}$$



$$E = hf - \phi$$

At $\phi = 0$, $E = hf$, the gradient $= \frac{f}{E}$, therefore $h = \frac{1}{\text{gradient}}$

$$\text{gradient} = 1.5058 \times \frac{10^{14}}{10^{-19}} = 1.5058 \times 10^{33}$$

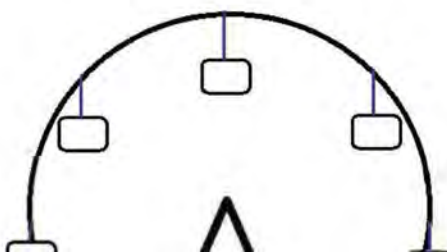
$$h = \frac{1}{\text{gradient}} = 6.640988179 \times 10^{-34}$$

$$\text{Percentage error} = \frac{6.64098819 \times 10^{-34} - 6.626 \times 10^{-34}}{6.626 \times 10^{-34}} = 0.23\% \text{ so it is very accurate}$$

Question 26 (9 marks)

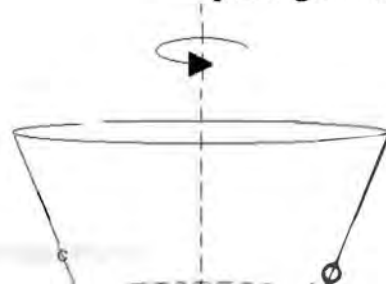
The picture shows two rides at a theme park, the “Ferris wheel” and the “Spinning Cone”. In motion, both rides move at constant speed.

Ferris wheel



47

Spinning Cone



- a) Discuss the total energy and work done on a passenger in the Ferris wheel and the Spinning cone. In your answer, discuss how gravitational potential energy, kinetic energy, and work done by forces change during the motion. (4)

Marking Criteria	Marks
Provides a thorough discussion including gravitational potential energy, kinetic energy, and work done by forces during the motion.	4
Provides a sound discussion including gravitational potential energy, kinetic energy, and work done by forces during the motion.	3
Provides outlines of gravitational potential energy, kinetic energy, and work done by forces during the motion	2
Provide some relevant information	1

Sample answer

In a Ferris wheel, the passenger moves in a vertical circle, so their height above the ground is constantly changing. As the passenger ascends, their gravitational potential energy (U) increases, thus KE decreases, and as they descend, U decreases and KE increases. As the Ferris wheel rotates at a constant speed, the kinetic energy remains constant throughout the motion. Since the passenger returns to the same height and speed after one complete revolution, there is no net change in mechanical energy. Therefore, the total work done on the passenger by gravity is zero. Although gravity does positive work as the passenger moves downward and negative work as they move upward, these effects cancel out over a full rotation

In contrast, a spinning cone moves the passenger in a horizontal circle at a constant height. In this case, gravitational potential energy remains unchanged because there is no change in vertical position. The centripetal force required for circular motion acts toward the centre of the circle, perpendicular to the direction of motion. Since work is only done when a force has a component in the direction of motion, this force does no work on the passenger. As a result, if the ride spins at a constant speed, the kinetic energy remains constant and no net work is done on the passenger.

- b) During the ride on the spinning cone, the cone-shaped enclosure spins about a vertical axis, causing the passenger to rotate in a horizontal circle at a height of 20.0 metres. The coefficient of static friction between the wall and the passenger is 0.650.

Determine the minimum and maximum speeds required so that the passenger does not slip down or up the wall, respectively. (5)

Marking Criteria	Marks
Provides complete steps with derivations to calculate the minimum and maximum velocity, justification for why there is no $V_{\max}$	5

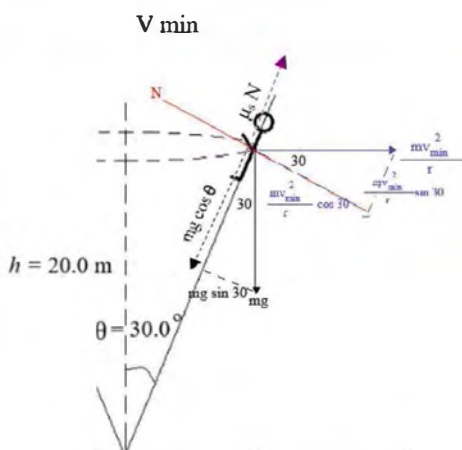
Provides complete steps with derivations to calculate the minimum and maximum velocity	4
- Provides complete steps to calculate the minimum and maximum velocity, justification for why there is no Vmax OR - Provides complete steps with derivation to calculate the minimum and maximum velocity	3
Provides complete steps to calculate the minimum and maximum velocity	2
Provide some relevant information	1

Sample answer

The cone is spinning and will have a centripetal force towards the centre of the cone acting on the passenger

From the frame of reference of the passenger, for the passenger to remain at 20 m there is an equal but opposite force to the centripetal force of the cone in the Normal component based on Newton's 3rd Law.

$$\tan 30 = \frac{r}{20}, \text{ so } r = 20 \tan 30$$



$$N = mg \sin 30 + \frac{mv^2}{r} \cos 30$$

$$\mu_s N + \frac{mv^2}{r} \sin 30 = mg \cos 30$$

Sub N into 2nd eqn

$$\mu_s \left(mg \sin 30 + \frac{mv^2}{r} \cos 30 \right) + \frac{mv^2}{r} \sin 30 = mg \cos 30$$

$$\mu_s g \sin 30 + \mu_s \frac{v^2}{r} \cos 30 + \frac{v^2}{r} \sin 30 = g \cos 30$$

$$\mu_s \frac{v^2}{r} \cos 30 + \frac{v^2}{r} \sin 30 = g \cos 30 - \mu_s g \sin 30$$

$$\frac{v^2}{r} (\cos \mu_s \cos 30 + \sin \sin 30) = g \cos 30 - \mu_s \sin \sin 30$$

$$v = \sqrt{\frac{rg \cos 30 - \mu_s \sin \sin 30}{\cos(\mu_s \cos 30 + \sin \sin 30)}} = \sqrt{\frac{20 \tan 30 \times 9.830 - 0.650 \sin \sin 30}{30 + \sin \sin 30}}$$

$$v = 31.21 \text{ ms}^{-1}$$

V max

$h = 20.0 \text{ m}$

$\theta = 30.0^\circ$

$$N = mg \sin \sin 30 + \frac{mv^2}{r} \cos \cos 30$$

$$\frac{mv^2}{r} \sin \sin 30 = \mu_s N + mg \cos \cos 30$$

Sub N into 2nd eqn

$$\frac{mv^2}{r} \sin \sin 30 = \mu_s (mg \sin \sin 30 + \frac{mv^2}{r} \cos \cos 30) + mg \cos \cos 30$$

$$\frac{v^2}{r} \sin \sin 30 = \mu_s g \sin \sin 30 + \mu_s \frac{v^2}{r} \cos \cos 30 + g \cos \cos 30$$

$$\frac{v^2}{r} \sin \sin 30 - \mu_s \frac{v^2}{r} \cos \cos 30 = \mu_s g \sin \sin 30 + g \cos \cos 30$$

$$\frac{v^2}{r} (30 - \mu_s \cos \cos 30) = g (\mu_s \sin \sin 30 + \cos \cos 30)$$

$$v = \sqrt{\frac{rg (\mu_s \sin \sin 30 + \cos \cos 30)}{30 - \mu_s \cos \cos 30}}$$

However, $30 - \mu_s \cos \cos 30 = 30 - 0.65 \cos \cos 30$

$= -0.063$

The denominator cannot be -ve, therefore there is no max. The passenger will never slip up.

Question 27 (4 marks)

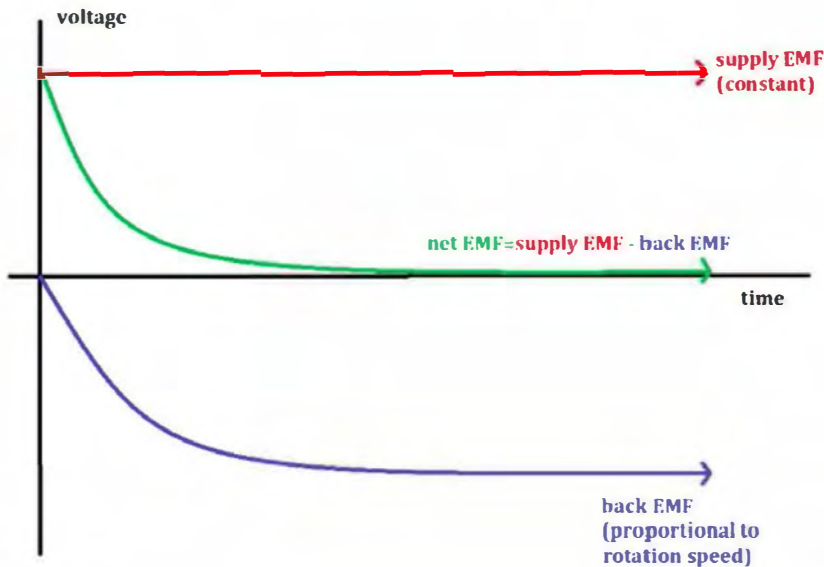
Explain the relationship between the supplied emf, back emf, and net emf in a motor as it starts up and continues to spin. Support your answer with a labelled graph. (4)

Marking Criteria	Marks
• Provides a thorough explanation with a graph	4
• Provides a sound explanation with a graph	3
• Draw a graph to show the relationship	2
OR	
• Provides an explanation	
• Provide some relevant information	1

Sample answer

When a motor is first turned on, the supplied emf (from the power source) is at its maximum because the motor is not yet spinning. At this moment, the back emf is zero because there is no rotation and hence the coil doesn't experience any change in flux, so the back emf is zero. Therefore, the net emf (which drives the current) is equal to the supplied emf.

As the motor starts rotating, the coil experiences a faster rate of change of flux and hence induces a larger back emf, thereby resulting in a decrease in net emf as $\text{net EMF} = \text{supply emf} - \text{back emf}$ based on conservation of energy. This continues until the net emf approaches zero (in an ideal motor), thereby producing a graph like the one shown below.



Question 28 (4 marks)

Discuss the impacts Einstein's theory of Special Relativity will have on space travel.

(4)

Marking Criteria	Marks
Provides a thorough discussion of the impacts that Einstein's theory of Special Relativity will have on space travel	4
Provides a sound discussion of the impacts that Einstein's theory of Special Relativity will have on space travel	3
Provides a basic discussion of the impacts that Einstein's theory of Special Relativity will have on space travel	2
Provide some relevant information	1

Sample answer

Einstein's Special Relativity has significant implications for space travel.

As a spacecraft approaches the speed of light, time dilation means time for astronauts passes more slowly than on Earth, so that they could complete journeys to distant stars within their lifetime, but much more extended periods would pass on Earth, creating challenges for communication and return missions.

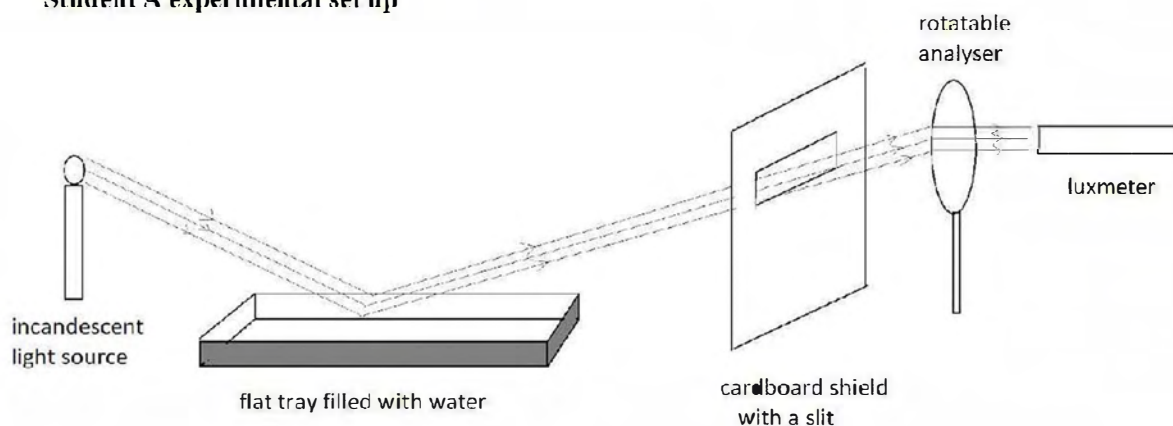
Length contraction also occurs, so distances in the direction of travel appear shorter to the astronauts, making interstellar travel seem more achievable from their perspective. However, the energy requirements grow enormously because, as speed increases, relativistic mass increases, making it practically impossible to reach the speed of light with current technology.

Relativity also creates communication issues, since signals still travel at light speed, and delays would become significant over interstellar distances.

Question 29 (6 marks)

The following diagram shows an experiment that investigates the nature of light.

Student A experimental set up



An initial reading of 2,000 lux was recorded. However, when the analyser was rotated 90° , the luxmeter recorded a reading of 7 lux.

Explain how the experiment supports Maxwell's theory of electromagnetism.

(6)

Marking Criteria	Marks
Provides an explanation with reference to polarisation of light, the lux meter reading and specific Maxwell's theory	6
Provides an explanation with reference to polarisation of light, the lux meter reading and/or specific Maxwell's theory	5
Provides an explanation with reference to polarisation of light and/or the lux meter reading and specific Maxwell's theory	4

• Outlines how it supports with reference to polarisation of light and/or the lux meter reading and specific Maxwell's theory	2-3
• Provide some relevant information	1

The experiment shows that when light is reflected off water, it becomes partially polarised. When it travels through a plane slit, the lux meter is reduced to 7 lux, showing that light has transverse oscillations rather than longitudinal ones. This supports Maxwell's theory because his equations predicted that electromagnetic waves, including light, consist of oscillating electric and magnetic fields that are perpendicular to each other and the direction of travel—a transverse wave property. Since only transverse waves can be polarised, the observation of polarisation is strong evidence that light behaves exactly as Maxwell's theory described.

Question 30 (7 marks)

Explain how the study of circular motion and waves helped scientists develop and change the models of the atom over time. (7)

Marking Criteria	Marks
• Provides a thorough explanation of how circular motion and waves helped scientists develop and change the model from Bohr-Rutherford to Schrödinger	7
• Provides a sound explanation of how circular motion and waves helped scientists develop and change the model from Bohr-Rutherford to Schrödinger	6
• Provides a thorough explanation of how circular motion and waves helped scientists develop and change the model from Bohr-Rutherford to de Broglie	4-5
• Provides a sound explanation of how circular motion and/or waves helped scientists develop and change the model from Bohr-Rutherford to Schrödinger	2-3
• Provide some relevant information	1-2

Sample answer

The development of atomic models has been deeply influenced by the study of circular motion and wave behaviour, especially during the transition from classical to quantum physics.

In 1913, Niels Bohr applied principles of circular motion to Rutherford's nuclear model by proposing that electrons travel in fixed, quantised circular orbits around the nucleus. He used the concept of centripetal force provided by electrostatic attraction and introduced a key quantum condition: the angular momentum of an electron is quantised as

$L = n\hbar$. This explained the discrete spectral lines of hydrogen, as energy is only absorbed or emitted when electrons transition between allowed orbits.

However, Bohr's model still relied on classical physics and could not fully explain atoms with more than one electron. This limitation was addressed through the study of wave-particle duality. In 1924, Louis de Broglie proposed that electrons, like photons, have wave properties, with a wavelength $\lambda = \frac{h}{p}$. This explained why only certain orbits were allowed—those where the electron's matter wave could form a stable standing wave.

Experimental support came from the Davisson-Germer experiment, which demonstrated electron diffraction, confirming that electrons exhibit wave-like behaviour. These discoveries led to the abandonment of the idea of fixed electron paths.

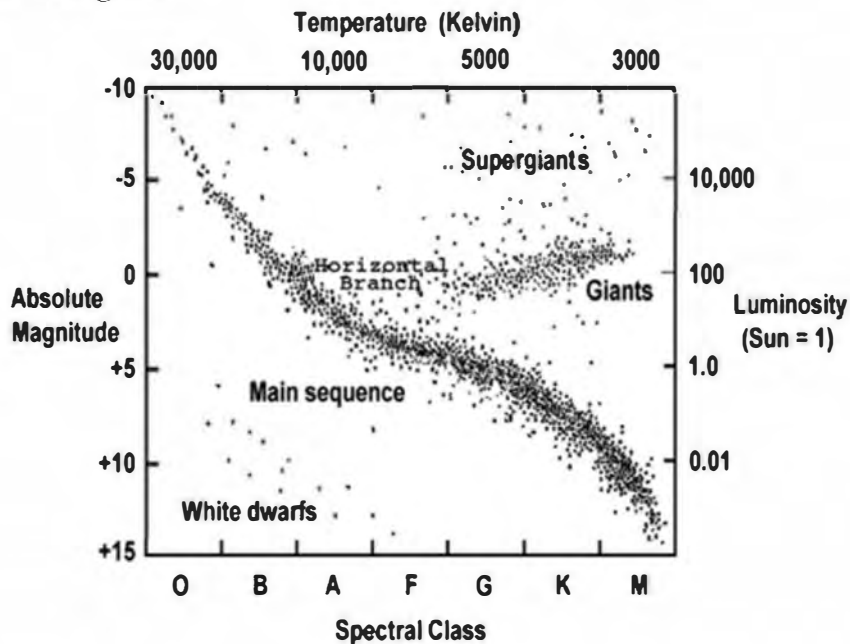
In 1926, Erwin Schrödinger developed a wave equation that treated electrons as wavefunctions. This quantum mechanical model replaced orbits with orbitals—three-dimensional probability distributions where electrons are likely to be found. It provided a more accurate and general description of atomic structure, especially for multi-electron systems.

In summary, circular motion provided an early framework for atomic structure through quantised orbits, while the wave nature of electrons transformed our understanding, leading to the probabilistic and mathematically robust quantum model of the atom.

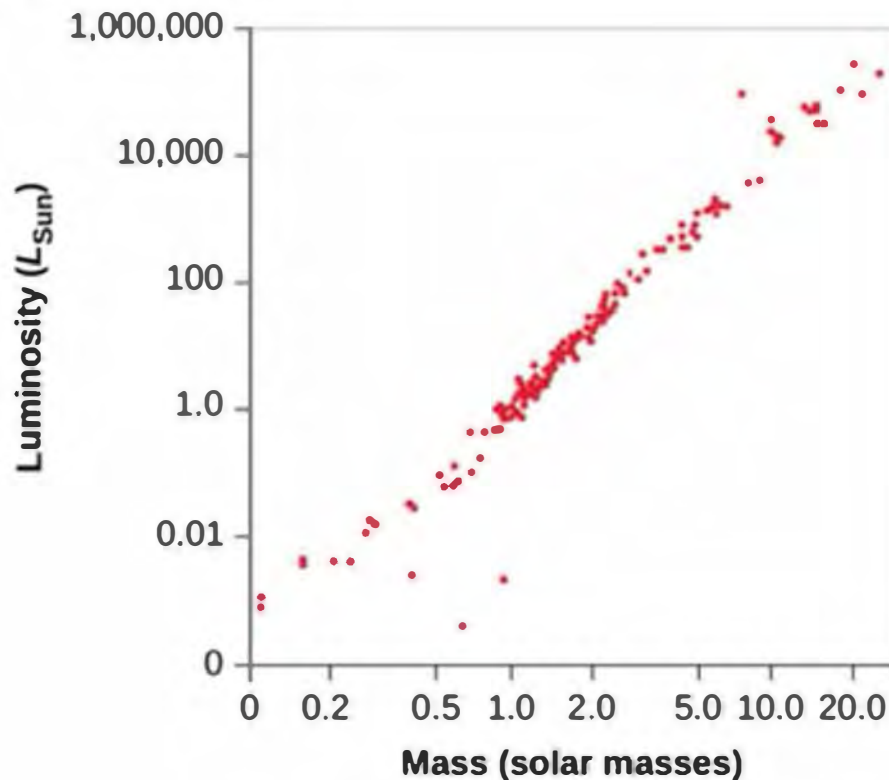
Question 31 (8 marks)

Scientists frequently utilise graphical relationships to model systems, explain underlying mechanisms, and make predictions about natural phenomena. For example, by interpreting stellar spectra, we know our Sun is a G star with a surface temperature of 5,772 K.

Graph 1- H-R Diagram



Graph 2- Luminosity Vs Mass

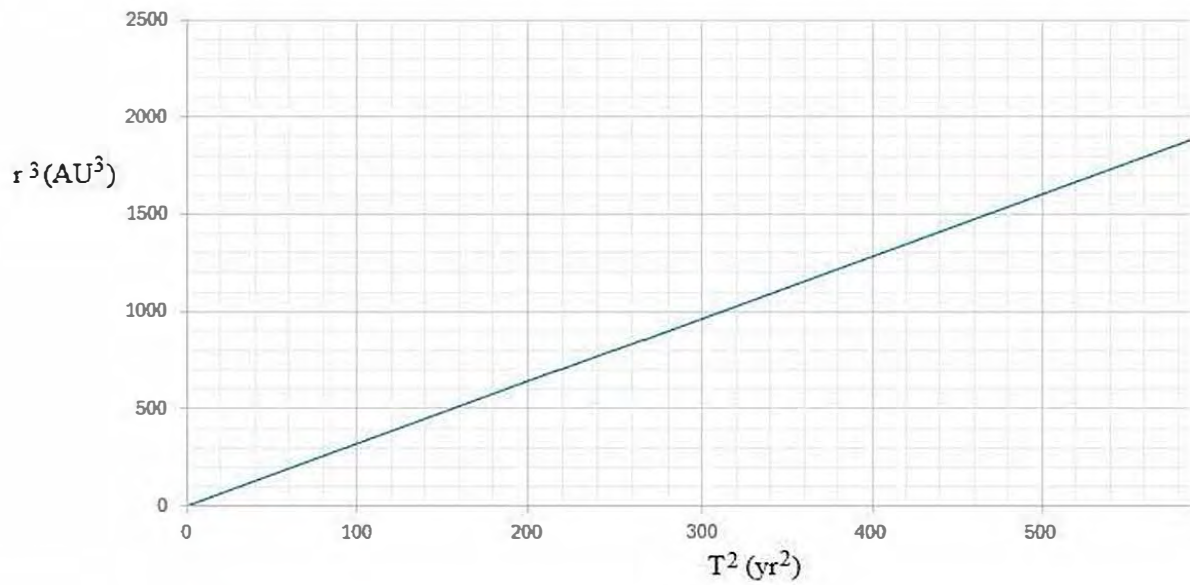


Mass of the Sun = 1.989×10^{30} kg

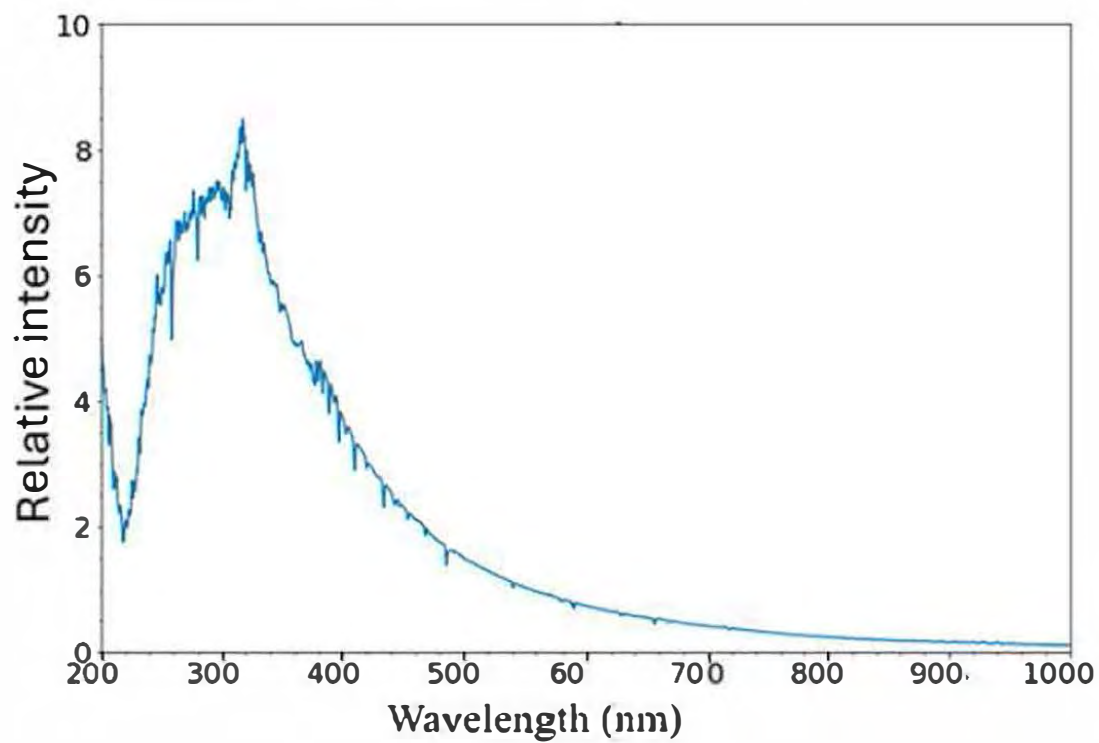
Question 31 continues on page 26

Question 31 continued

Graph 3 – R^3 vs T^2 for satellites orbiting Star X



Graph 4- Relative intensity vs wavelength for Star X



Question 31 continues on page 27

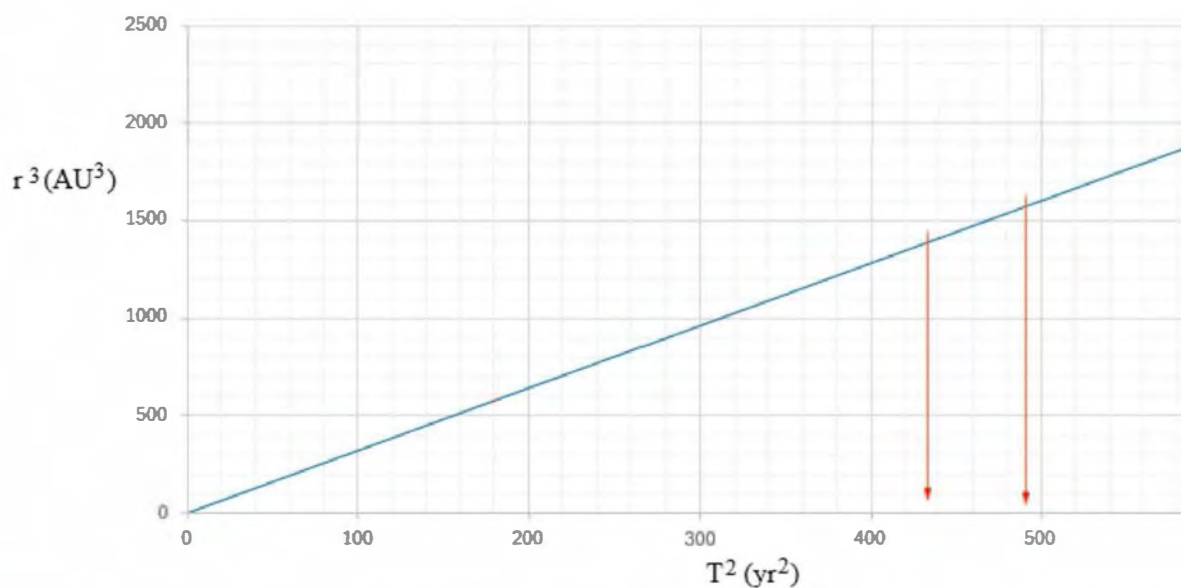
Question 31 continued

By interpreting these graphical relationships in combination, describe how the physical and chemical properties of Star X differ from those of our Sun, including their chemical processes. (8)

Marking Criteria	Marks
Provide a thorough analysis of all four graphs individually and together, explain the difference in the physical and chemical properties of Star X and the Sun, including their chemical process	8
Provide a sound analysis of all four graphs individually and together, explain the difference in the physical and chemical properties of Star X and the Sun, including their chemical process	6-7
Provide a sound analysis of the graphs individually and together, explain the difference in the physical and chemical properties of Star X and the Sun, including their chemical process	4-5
Provide a sound analysis of the graphs, explain the difference in the physical and chemical properties of Star X and the Sun	2-3
Provide some relevant information	1

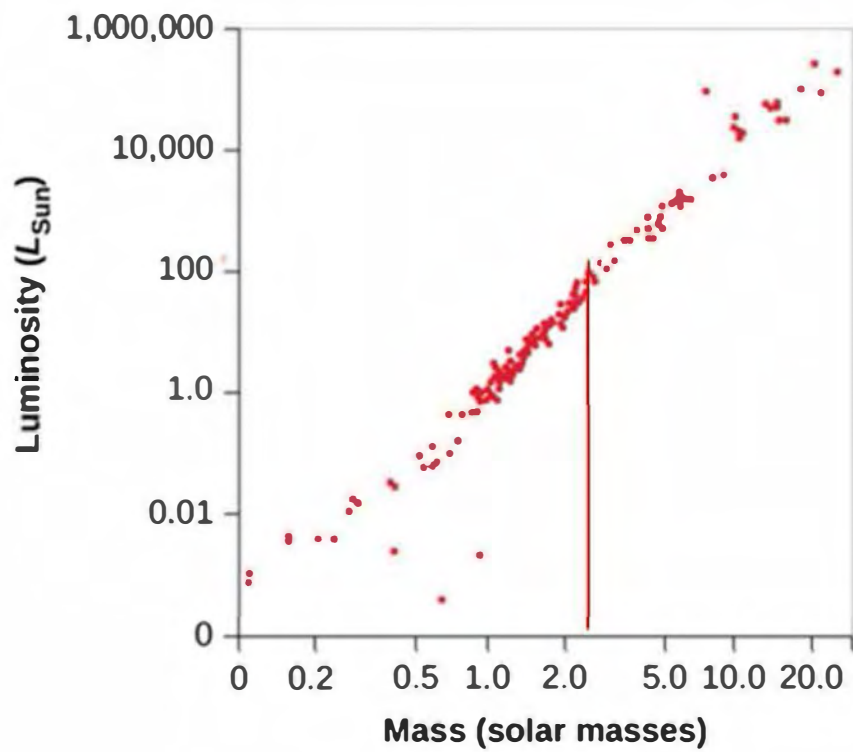
Sample answer

From the Graph 3 – R^3 vs T^2 for satellites orbiting Star X



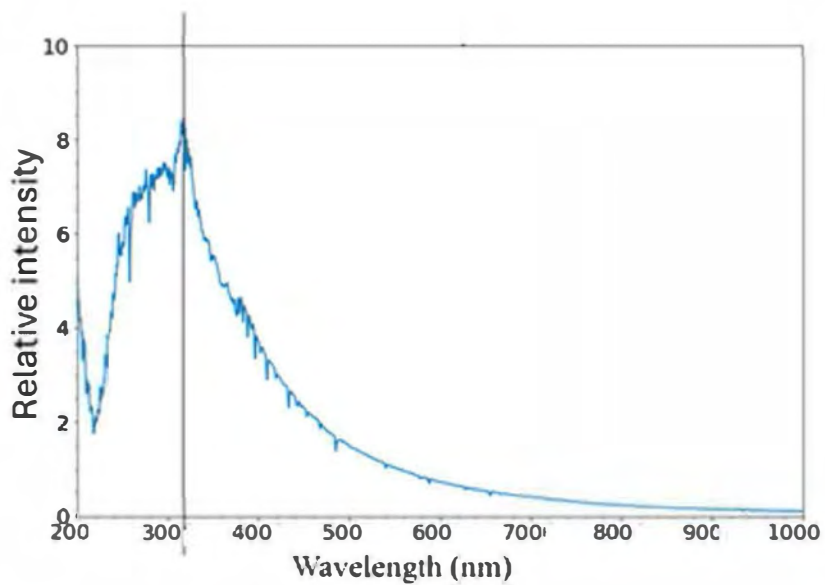
Based on the gradient $= \frac{200}{60} = 3.3 M_o = 3.3 \times \text{the mass of our sun}$

From Graph 2- Luminosity Vs Mass



Since Star X is 3.3 times the mass of the Sun, its luminosity is 100 times that of our Sun.

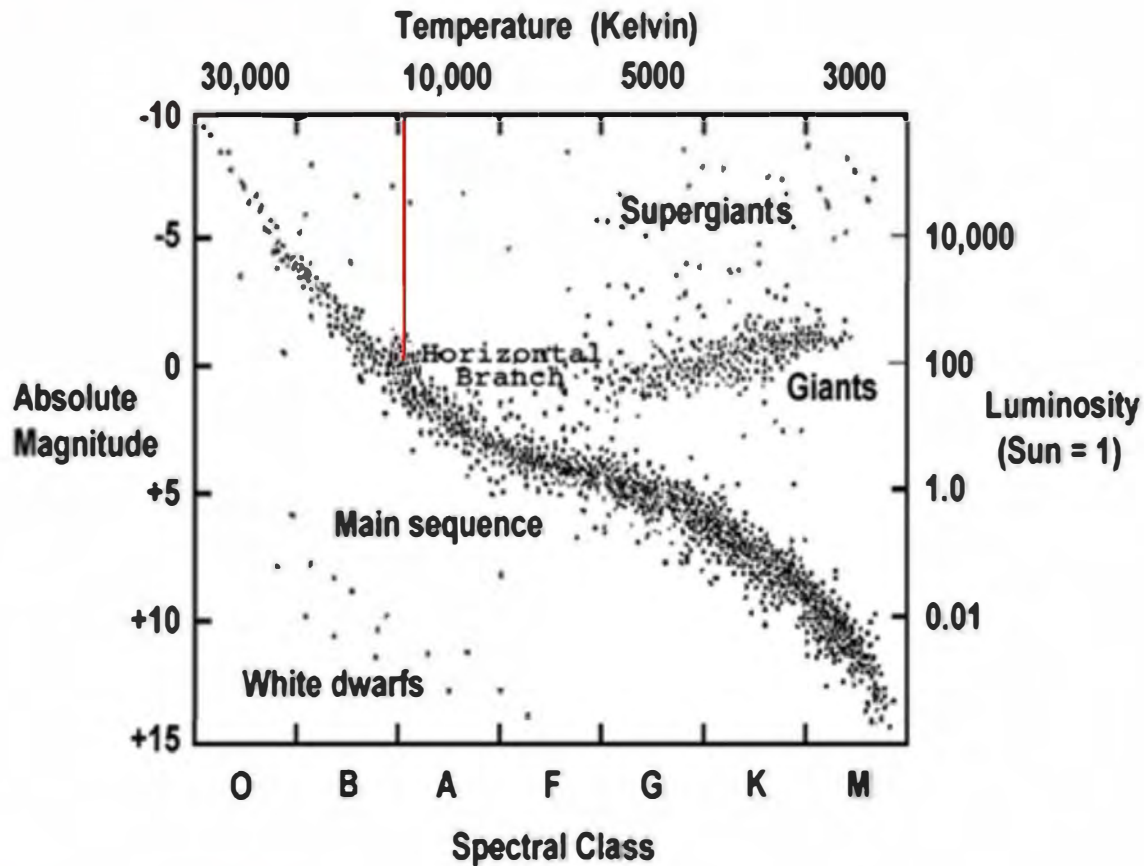
From the Graph 4- Relative intensity vs wavelength for Star X



Wavelength max. peak = 3200 nm

$$\lambda = \frac{b}{T}, \text{ so } T = \frac{b}{\lambda} = \frac{2.898 \times 10^{-3}}{3200 \times 10^{-9}} = 9056 \text{ K as compared to our Sun, which is } 5772 \text{ K}$$

From Graph 1- H-R Diagram



Main sequence A-class star.

Overall:

Property	Star X (A-class, Luminosity = 100 $L_{\odot}$)	Sun (G-class, Luminosity = 1 $L_{\odot}$)
Spectral Type	A-type main sequence star	G-main sequence star
Surface Temperature	~10,000 K	~5,772 K
Colour	White or bluish white	Yellow

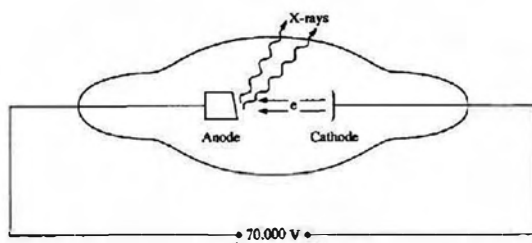
Star X is significantly hotter and emits more high-energy light (especially UV), whereas the Sun emits more in the visible spectrum, making it cooler and dimmer. Star X is larger and more massive, contributing to its higher pressure and temperature in the core.

Both stars are made mostly of hydrogen and helium. However, Star X will undergo more CNO cycles than the Sun, which is predominantly the proton-proton chain, which is dominant in stars of $1 M_{\odot}$ or less. Star X shows weaker metal absorption lines because of higher ionisation, while the Sun shows stronger metal lines due to its cooler atmosphere.

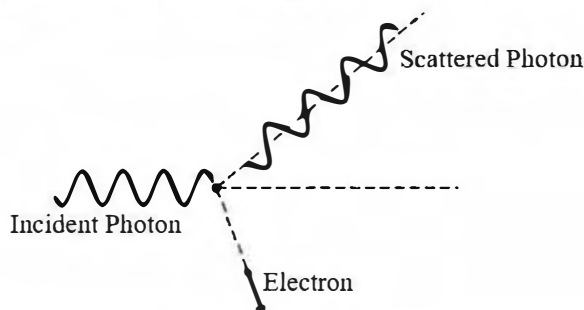
Star X has a much shorter life due to rapid hydrogen burning. The Sun is more stable and long-lived.

Question 32 (5 marks)

In the X-ray tube shown above, a potential difference of 70,000 volts is applied across the two electrodes. Electrons emitted from the cathode are accelerated to the anode, where X-rays are produced.



An X-ray photon of the maximum energy produced by this tube leaves the tube and collides elastically with an electron at rest. As a result, the electron recoils and the X-ray is scattered, as shown above. The frequency of the scattered X-ray photon is 1.64×10^{19} hertz.



Determine the de Broglie wavelength of the electron.

(5)

Marking Criteria	Marks
Calculate the de Broglie wavelength of the electron with reference to conservation of energy	5
Calculate the de Broglie wavelength of the electron	4
Provides steps to calculate the de Broglie wavelength of the electron with reference to conservation of energy	3
	2
Provide a relevant step	1

The energy of the incident X-ray photon is equal to the energy of the electron accelerated from the cathode to the anode

$$E = Vq = E_{\text{photon}(i)} \text{ , therefore } E_{\text{photon}(i)} = (70000)(1.602 \times 10^{-19}) = 1.121 \times 10^{-14}$$

From energy conservation, the total energy before must equal the energy after

$$E_{\text{photon}(i)} = E_{\text{photon}(f)} + E_{\text{electron}}$$

$$hf_i = hf_f + K_e$$

$$(70000)(1.602 \times 10^{-19}) = (6.626 \times 10^{-34})(1.64 \times 10^{19}) + K_e$$

$$K_e = 3.408 \times 10^{-16} \text{ J}$$

Since $\lambda = \frac{h}{mv} = \frac{h}{p}$ The energy of the recoiled electron can help us find its momentum. Using $K = \frac{1}{2} mv^2$
 $3.408 \times 10^{-16} \text{ J} = \frac{1}{2} 9.109 \times 10^{-31} \text{ v}^2$ so $v = 27354541 \text{ m/s}$
 $p = (9.108 \times 10^{-31})(2.7 \times 10^7) = 2.49145164 \times 10^{-23} \text{ kg-m/s}$
 $p = h / \lambda$ that is $2.49145164 \times 10^{-23} = (6.626 \times 10^{-34}) / \lambda$ therefore
 $\lambda = 2.65949372 \times 10^{-11} \text{ m}$